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Magm functions

Synopsis

The functions prefixed with Magm make up the Advanced Geometric Matcher (AGM) module. The AGM module can train models from images and find occurrences of models in a target image using their extracted edges. The module can help solve machine vision problems, such as alignment, measurement, and inspection.

Functions

MagmAlloc
MagmAllocResult
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MagmPreprocess
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MagmSave
MagmStream
MagmTrain

MagmAlloc

Synopsis

Allocate a find or train AGM context.

Syntax

```
AIL_ID MagmAlloc(
    AIL_ID SysId,          //in
    AIL_INT64 ContextType, //in
    AIL_INT64 ControlFlag, //in
    AIL_ID *ContextIdPtr   //out
)
```

Description

This function allocates a find or train AGM context on the specified system. A find AGM context contains the information necessary to perform an MagmFind() operation, while a train AGM context contains the information necessary to perform an MagmTrain() operation.

Define and add models to an AGM context using MagmDefine(). You add a single-definition model directly to a find AGM context using MagmDefine(). To add a composite-definition model to a find AGM context, you must first add the model to a train AGM context using MagmDefine(), train the model using MagmTrain(), and then copy the trained composite-definition model to the find AGM context using MagmCopyResult(). You can adjust individual model settings using MagmControl().

When the AGM context is no longer required, release it using MagmFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the AGM context's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the context. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextType

Specifies the type of AGM context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the type of context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GLOBAL_EDGE_BASED_FIND	Specifies to allocate a find AGM context. This context contains settings and the model to find with MagmFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_EDGE_BASED_TRAIN	Specifies to allocate a train AGM context. This context contains settings and the composite-definition model to train with MagmTrain().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

ContextIdPtr

Specifies the address of the variable in which to write the AGM context identifier or specifies the data type that the function should use to return the AGM context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_AGM_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the AGM context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find AGM context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated find AGM context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the train AGM context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated train AGM context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the AGM context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_AGM_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmAllocResult

Synopsis

Allocate a find or train AGM result buffer.

Syntax

```
AIL_ID MagmAllocResult(
    AIL_ID SysId,          //in
    AIL_INT64 ResultType,  //in
    AIL_INT64 ControlFlag, //in
    AIL_ID *ResultAgmIdPtr //out
)
```

Description

This function allocates a find or train AGM result buffer on the specified system. A find AGM result buffer stores results obtained from an MagmFind() operation, while a train AGM result buffer stores results obtained from an MagmTrain() operation.

When the AGM result buffer is no longer required, release it using MagmFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the AGM result buffer's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of AGM result buffer. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the type of results buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_GLOBAL_EDGE_BASED_FIND_RESULT	Specifies to allocate a find AGM result buffer. This result buffer holds the results produced from calling MagmFind().	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_EDGE_BASED_TRAIN_RESULT	Specifies to allocate a train AGM result buffer. This result buffer holds the results produced from calling MagmTrain().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

ResultAgmIdPtr

Specifies the address of the variable in which to write the AGM result buffer identifier. Since the **MagmAllocResult**() function also returns the AGM result buffer identifier, you can set this parameter to M_NULL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_AGM_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the AGM result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find AGM result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated find AGM result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the train AGM result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated train AGM result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the AGM result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_AGM_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmControl

Synopsis

Control a model in an AGM context or a train AGM result buffer setting.

Syntax

```
void MagmControl(
    AIL_ID ContextOrResultId, //out
    AIL_INT64 Index,          //in
    AIL_INT64 ControlType,    //in
    AIL_DOUBLE ControlValue   //in
)
```

Description

This function allows you to control a setting of a single-definition or composite-definition model in a find AGM context, of a composite-definition model in a train AGM context, or of a train AGM result buffer. These settings control the execution of MagmFind() or MagmTrain() operations. Most of the control type settings can be inquired using MagmInquire().

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MagmControl()** internally calls the *AIL_DOUBLE* version of this function (MagmControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MagmControlInt64()).

Parameters

ContextOrResultId

Specifies the identifier of a find or train AGM context or train AGM result buffer to control.

See the Parameter associations section for possible values that can be returned.

Index

Specifies what to control. Set this parameter to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a model or result buffer																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_AGM_MODEL_INDEX(AIL_INT ModelIndex)	Specifies to control an individual model in an AGM context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ModelIndex Set this parameter to the following:															

			0 Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to control a general setting of an AGM result buffer, if one is specified.			c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters and possible values returned to the **ContextOrResultId** parameter.

- For controlling a model in an AGM context
- For controlling a train AGM result buffer

The following ContextOrResultId, ControlType, and ControlValue parameter settings can be specified for models in AGM contexts. In this case, you must set the Index parameter to the index of a specific model, using M_AGM_MODEL_INDEX().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a model in an AGM context																				
ContextOrResultId - Possible values returned			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlType																		
		ControlValue																	
Find AGM context ID with a composite-definition model			Specifies a find AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND and used in MagmFind() operations. It must contain a trained composite-definition model.																

M_ACCEPTANCE_COVERAGE	Sets the minimum coverage score required for an occurrence to be considered a match. The coverage score is the percentage of the total length of the model's edgels found in the occurrence. 100% indicates that for each of the model's edgels, a corresponding edgel was found in the occurrence. Note, a model's edgel corresponds to an edgel in the occurrence if the distance between them is less than or equal to M_MAX_ASSOCIATION_DISTANCE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable coverage score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_DETECTION	Sets the minimum detection score required for an occurrence to be considered a match. The detection score represents how confident you should be that Aurora Imaging Library found a true occurrence. Note, this value is only used if M_ACCEPTANCE_DETECTION_SOURCE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable detection score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_DETECTION_SOURCE	Sets whether to use the trained acceptance level or a custom acceptance level for the detection score. You can typically start with the acceptance level established during training and specify a higher acceptance level if the search speed is too slow, or if the search finds false occurrences. You should specify a lower acceptance level if the search does not find all model occurrences. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_FROM_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the detection score's acceptance level established during the training of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the value set with M_ACCEPTANCE_DETECTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT		Sets the minimum fit score required for an occurrence to be considered a match. The fit score is a measure of the correlation of the edgels in the model to their corresponding edgels in the occurrence. It is based on the distance between the model's edgels and the occurrence's edgels. Note, only those model edgels that are less than or equal to M_MAX_ASSOCIATION_DISTANCE from an occurrence edgel are used in the fit score calculation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT_OVERALL		Sets the minimum overall fit score required for an occurrence to be considered a match. The overall fit score is a measure of the correlation of all edgels in the model to those of the occurrence. It is based on the distance between the model's edgels and the occurrence's edgels. Note, all model edgels are used in the overall fit score calculation, including those without a corresponding edgel in the occurrence. This control type is only available when M_MODEL_SOURCE is set to M_USER_IMAGE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b

		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable overall fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_DELTA_ANGLE_SOURCE		Sets whether to use the trained detection angle range or a custom detection angle range to limit the angle of detection. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_FROM_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the detection angle range established during the training of the model. If M_TRAIN_DETECTION_ANGLE was set to M_ENABLE, the detection angle range is 360.0°. If M_TRAIN_DETECTION_ANGLE was set to M_DISABLE, the detection angle range is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the values set with M_DETECTION_DELTA_NEG_ANGLE and M_DETECTION_DELTA_POS_ANGLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DETECTION_DELTA_NEG_ANGLE	Sets the lower limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *LowerLimit* = M_DETECTION_REF_ANGLE - M_DETECTION_DELTA_NEG_ANGLE. The detection angle range should be used to cover the expected variance in the angle of detection. Note that the detection angle range only limits the angle of detection. The angle at which an occurrence is detected might not be exact; its true angle can typically be within 10° of the angle of detection, even if that falls outside of the detection angle range. If M_FIT_MODE is disabled, the returned angle of an occurrence will always be the angle at which it was detected and will therefore always be within the detection angle range. If M_FIT_MODE is enabled, a fit operation is performed starting from the angle of detection. In refining the angle so that it is more accurate, the angle of the occurrence's reference axis returned by MagmGetResult() with M_ANGLE can end up being outside of the detection angle range. Note that the greater the range of angles to be searched, the slower the search. To set the upper limit for the detection angle range, use M_DETECTION_DELTA_POS_ANGLE. This control type is only used when M_DETECTION_DELTA_ANGLE_SOURCE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the lower limit of the detection angle range, in degrees. Note that M_DETECTION_DELTA_NEG_ANGLE + M_DETECTION_DELTA_POS_ANGLE cannot be greater than 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DETECTION_DELTA_POS_ANGLE	Sets the upper limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *UpperLimit* = M_DETECTION_REF_ANGLE + M_DETECTION_DELTA_POS_ANGLE. The detection angle range should be used to cover the expected variance in the angle of detection. Note that the detection angle range only limits the angle of detection. The angle at which an occurrence is detected might not be exact; its true angle can typically be within 10° of the angle of detection, even if that falls outside of the detection angle range. If M_FIT_MODE is disabled, the returned angle of an occurrence will always be the angle at which it was detected and will therefore always be within the detection angle range. If M_FIT_MODE is enabled, a fit operation is performed starting from the angle of detection. In refining the angle so that it is more accurate, the angle of the occurrence's reference axis returned by MagmGetResult() with M_ANGLE can end up being outside of the detection angle range. Note that the greater the range of angles to be searched, the slower the search. To set the lower limit for the detection angle range, use M_DETECTION_DELTA_NEG_ANGLE. This control type is only used when M_DETECTION_DELTA_ANGLE_SOURCE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the upper limit of the detection angle range, in degrees. Note that M_DETECTION_DELTA_NEG_ANGLE + M_DETECTION_DELTA_POS_ANGLE cannot be greater than 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b

M_DETECTION_MULTI_ANGLE_MODE		Sets the mode with which to detect occurrences when searching at multiple angles. The M_GLOBAL mode can accelerate search speed, but can also increase the risk of missing overlapping occurrences that have close centers. You should specify M_INDIVIDUAL if the search is not finding all model occurrences, but note that this will slow down the search. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_GLOBAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GLOBAL	Specifies to detect occurrences across all search angles globally.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INDIVIDUAL	Specifies to detect occurrences at each search angle individually.	c		e	f	g	a		i		j	k	l	m	n	b
M_DETECTION_REF_ANGLE		Sets the nominal search angle; this is the angle at which you expect to find the model's reference axis in the target. Note that occurrences can be slightly rotated from the search angle and still be detected. The exact degree varies depending on your model, but the MagmFind() operation will typically detect occurrences within 10° of the search angle. If you expect greater degrees in variation, you should specify a detection angle range, using M_DETECTION_DELTA_NEG_ANGLE and M_DETECTION_DELTA_POS_ANGLE. Note that specifying a nominal search angle does not decrease the search speed. This control type is only used when M_DETECTION_DELTA_ANGLE_SOURCE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* < 360.0	Specifies the nominal search angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_DETECTION_STEP_ANGLE		Sets the step angle at which to search for occurrences of the model within the specified detection angle range. Note that the smaller the step angle, the slower the search. The search is performed at the nominal search angle and at every +/- M_DETECTION_STEP_ANGLE degrees within the detection angle range. Note that occurrences can be slightly rotated from the search angle and still be detected. You should typically set the step angle to this rotation * 2. For example, if your model can tolerate the occurrence being offset from its search angle by ±5°, you should set the step angle to 10°. This control type is only used when M_DETECTION_DELTA_ANGLE_SOURCE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
	5.0 <= *Value* <= 180.0	Specifies the step angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_MODE		Sets the mode of the fit operation. The fit mode affects positional accuracy and search speed. The positional results are most accurate when the fit operation uses subpixel precision, but this can reduce the speed of the search. Note that the fit can only be calculated with subpixel precision when you have explicitly specified an image of the model to use for the fit calculations (M_MODEL_IMAGE) and the edges of the target images are extracted (M_USE_MAGNITUDE_TARGET is disabled). You can disable the fit operation to speed up the search, in which case Aurora Imaging Library uses the features of the model and target to find occurrences, but it doesn't do any fitting. This results in a lower accuracy because the positional results are simply the positions of detection. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_PIXEL_PRECISION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the fit operation; the positional results are determined by detection only.	c		e	f	g	a		i		j	k	l	m	n	b

		M_PIXEL_PRECISION	Specifies to calculate the fit with pixel precision.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SUBPIXEL_PRECISION	Specifies to calculate the fit with subpixel precision. This value is only available when M_USE_MAGNITUDE_TARGET is set to M_DISABLE and M_MODEL_SOURCE is set to M_USER_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_ACCEPTED_OVERLAP		Sets the maximum allowable overlap between found occurrences, as a percentage. Note that M_MAX_ACCEPTED_OVERLAP is applied after possible occurrences are sorted. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 20.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the acceptable overlap, as a percentage. If set to 100%, found occurrences can overlap completely. If set to 0%, no found occurrence can overlap another by any amount.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_ASSOCIATION_DISTANCE		Sets the maximum distance between a model edgel and the nearest occurrence edgel. Model edgels falling outside the specified maximum distance are not used to perform the occurrence's regular fit operation, but will decrease the overall fit score and the coverage score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b
		1.0 <= *Value* <= 4.0	Specifies the maximum association distance, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_IMAGE	Sets the image buffer that contains the image of the model to use for the fit and coverage calculations. The model image should have a prototypical appearance with well-defined edges. You can prepare the image manually or define a child buffer within a training image. You must specify a model image when M_MODEL_SOURCE is set to M_USER_IMAGE. Specifying a prototypical image of the model allows for higher fit and coverage scores when model appearances are varied and the trained composite-definition model contains more edgels than a typical appearance. INQ	c		e	f	g	a		i		j	k	l	m	n	b
ModelImageId	Specifies the identifier of the image buffer containing the model image. Note that this buffer must be a 1-band, 8-bit unsigned image buffer. This image buffer must be the same size as the trained composite-definition model. You can inquire the size of the trained composite-definition model, using MagmInquire() with M_SIZE_X and M_SIZE_Y.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_SOURCE	Sets whether to use the trained composite-definition model or a prototypical image of the model for the fit and coverage calculations. You should use the trained composite-definition model if the labeled model appearances in your training images are uniform. When model appearances are varied, the trained composite-definition model contains additional edges. You can view the edges of the trained composite-definition model using MagmDraw() with M_DRAW_TRAINED_MODEL. If the trained composite-definition model contains a lot of edges, you should choose a model image for the fit and coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_FROM_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
M_FROM_TRAIN	Specifies to use the trained composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_IMAGE	Specifies to use the image set with M_MODEL_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PERSEVERANCE_DETECTION	Sets the algorithm's perseverance when searching for occurrences. This affects the number of candidates the algorithm attempts to validate. Increasing the perseverance will increase robustness, but will increase search time. You should try increasing the perseverance if the search is not finding all model occurrences, or if the target is complex and decreasing M_STEP_X and M_STEP_Y is insufficient. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_X	Sets the X-coordinate of the origin of the model's reference axis, relative to the model origin. Position results return the X-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the default X-coordinate will be used. For a composite-definition model, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-coordinate, in pixels. This value can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Y	Sets the Y-coordinate of the origin of the model's reference axis, relative to the model origin. Position results return the Y-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies that the default Y-coordinate will be used. For a composite-definition model, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-coordinate, in pixels. This value can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS		Sets the degree of smoothness (noise reduction) applied to the model image and target image(s) during the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. A setting of 0.0 indicates almost no noise reduction effect, while a setting of 100.0 indicates a very strong noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS_SOURCE		Sets whether to use the smoothness value set during training or a custom smoothness value for the degree of noise reduction applied to the model image and target image(s) during the edge extraction. You should customize the smoothness value if your training and target images have dissimilar noise levels. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_FROM_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the smoothness value set during the training of the model. This value was set with M_SMOOTHNESS in a train AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the smoothness value set with M_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b

M_SORT_CANDIDATES_SCORE		Sets the score used to sort candidates. Candidates are sorted in decreasing score order. The specified score affects which possible occurrences are filtered out. When two possible occurrences overlap above the value specified with M_MAX_ACCEPTED_OVERLAP, the possible occurrence with the lower score is removed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SCORE_DETECTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_COVERAGE	Specifies to use the coverage score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_DETECTION	Specifies to use the detection score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT	Specifies to use the fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT_OVERALL	Specifies to use the overall fit score to sort candidates. This value is only available when M_MODEL_SOURCE is set to M_USER_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
M_STABLE_EDGELS_MODE		Sets whether to use only stable edgels when searching for occurrences. A stable edgel is an edgel that remains persistent when extracting edges with slightly different smoothness values. This control type is only used and should only be enabled when M_MODEL_SOURCE is set to M_USER_IMAGE. Otherwise, an error is generated. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to use only stable edgels. All edgels are considered when searching for occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS_STABLE	Specifies to use only stable edgels. Unstable (noisy) edgels are ignored when searching for occurrences.	c		e	f	g	a		i		j	k	l	m	n	b

M_STEP_X		Sets the X step size of explored positions to search for occurrences. Increasing M_STEP_X and M_STEP_Y can accelerate search speed, but can also increase the risk of missing favorable occurrences. You can typically start with a value of 1 and increase M_STEP_X and M_STEP_Y if the speed is too slow. Increasing these values when you have a large model can decrease the search time while minimally affecting the occurrences' detection, fit, and coverage scores. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the X step size.	c		e	f	g	a		i		j	k	l	m	n	b
M_STEP_Y		Sets the Y step size of explored positions to search for occurrences. Increasing M_STEP_X and M_STEP_Y can accelerate search speed, but can also increase the risk of missing favorable occurrences. You can typically start with a value of 1 and increase M_STEP_X and M_STEP_Y if the speed is too slow. Increasing these values when you have a large model can decrease the search time while minimally affecting the occurrences' detection, fit, and coverage scores. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the Y step size.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_MODE		Sets the threshold mode of the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies no threshold. All edges are extracted.	c		e	f	g	a		i		j	k	l	m	n	b

		M_HIGH	Specifies a high threshold. M_HIGH should be used for images with some contrast variations, noise, and non-uniform illumination. M_HIGH always results in a lower (or equal) number of edgels than M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. All edges over a minimum noise-based estimated threshold are extracted. M_LOW always results in a lower (or equal) number of edgels than M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. M_MEDIUM should be used for multi-contrast images, or for images with a lot of noise or non-uniform illumination. M_MEDIUM always results in a lower (or equal) number of edgels than M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high threshold. Only the strongest edges in the image are extracted. M_VERY_HIGH always results in a lower (or equal) number of edgels than M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE_SOURCE		Sets whether to use the threshold mode set during training or a custom threshold mode for the edge extraction. You can use the threshold mode set during training if your training and target images have similar illumination and contrast. If your training and target images were captured in different conditions, the illumination and contrast might vary and you should customize the threshold mode accordingly. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_FROM_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the threshold mode set during the training of the model. This value was set with M_THRESHOLD_MODE in a train AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the threshold mode set with M_THRESHOLD_MODE.	c		e	f	g	a		i		j	k	l	m	n	b

Find AGM context ID with a single-definition model		Specifies a find AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND and used in MagmFind() operations. It must contain a single-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_COVERAGE	Sets the minimum coverage score required for an occurrence to be considered a match. The coverage score is the percentage of the total length of the model's edgels found in the occurrence. 100% indicates that for each of the model's edgels, a corresponding edgel was found in the occurrence. Note, a model's edgel corresponds to an edgel in the occurrence if the distance between them is less than or equal to M_MAX_ASSOCIATION_DISTANCE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable coverage score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_DETECTION	Sets the minimum detection score required for an occurrence to be considered a match. The detection score represents how confident you should be that Aurora Imaging Library found a true occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 60.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable detection score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACCEPTANCE_FIT	Sets the minimum fit score required for an occurrence to be considered a match. The fit score is a measure of the correlation of the edgels in the model to their corresponding edgels in the occurrence. It is based on the distance between the model's edgels and the occurrence's edgels. Note, only those model edgels that are less than or equal to M_MAX_ASSOCIATION_DISTANCE from an occurrence edgel are used in the fit score calculation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_FIT_OVERALL	Sets the minimum overall fit score required for an occurrence to be considered a match. The overall fit score is a measure of the correlation of all edgels in the model to those of the occurrence. It is based on the distance between the model's edgels and the occurrence's edgels. Note, all model edgels are used in the overall fit score calculation, including those without a corresponding edgel in the occurrence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable overall fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

	M_BOX_MARGIN_BOTTOM	Sets a margin at the bottom of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. The margins add a black space around the model, which affect whether two occurrences are considered overlapping. For example, even if none of an occurrence's edges touch another occurrence, if the margin around the bounding box of its edges overlaps with the margin and/or bounding box of another occurrence by any amount, the occurrences will be considered overlapping, and might be filtered out according to M_MAX_ACCEPTED_OVERLAP. Note that the margins will be included when drawing the model using MagmDraw(). The bounding box of the model's active edges plus the specified margins define the size of the model box. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

	M_BOX_MARGIN_LEFT	Sets a margin at the left side of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. The margins add a black space around the model, which affect whether two occurrences are considered overlapping. For example, even if none of an occurrence's edges touch another occurrence, if the margin around the bounding box of its edges overlaps with the margin and/or bounding box of another occurrence by any amount, the occurrences will be considered overlapping, and might be filtered out according to M_MAX_ACCEPTED_OVERLAP. Note that the margins will be included when drawing the model using MagmDraw(). The bounding box of the model's active edges plus the specified margins define the size of the model box. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

	M_BOX_MARGIN_RIGHT	Sets a margin at the right side of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. The margins add a black space around the model, which affect whether two occurrences are considered overlapping. For example, even if none of an occurrence's edges touch another occurrence, if the margin around the bounding box of its edges overlaps with the margin and/or bounding box of another occurrence by any amount, the occurrences will be considered overlapping, and might be filtered out according to M_MAX_ACCEPTED_OVERLAP. Note that the margins will be included when drawing the model using MagmDraw(). The bounding box of the model's active edges plus the specified margins define the size of the model box. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

	M_BOX_MARGIN_TOP	Sets a margin at the top of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. The margins add a black space around the model, which affect whether two occurrences are considered overlapping. For example, even if none of an occurrence's edges touch another occurrence, if the margin around the bounding box of its edges overlaps with the margin and/or bounding box of another occurrence by any amount, the occurrences will be considered overlapping, and might be filtered out according to M_MAX_ACCEPTED_OVERLAP. Note that the margins will be included when drawing the model using MagmDraw(). The bounding box of the model's active edges plus the specified margins define the size of the model box. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

	M_CAD_Y_AXIS	Sets the direction of the Y-axis for a model of type M_DXF_FILE. When the model is added to the AGM context from a CAD DXF file, the model will retain its coordinate system (the origin and the axes). Most CAD DXF files are oriented so that the Y-axis is positive going up, whereas the coordinate system Aurora Imaging Library uses is oriented with the Y-axis positive going down. So if you take the edge coordinates of an object from a CAD DXF file and put them in an image, the imaged object will look flipped when compared to the original object. This is also the case for a model defined from a CAD DXF file. This means that, unless the object is symmetrical, the match will not be made. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_FLIP.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FLIP	Specifies to flip the Y-axis for the model so that the Y-axis is positive going down.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NO_FLIP	Specifies not to flip the Y-axis for the model.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DETECTION_DELTA_NEG_ANGLE	Sets the lower limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *LowerLimit* = M_DETECTION_REF_ANGLE - M_DETECTION_DELTA_NEG_ANGLE. The detection angle range should be used to cover the expected variance in the angle of detection. Note that the detection angle range only limits the angle of detection. The angle at which an occurrence is detected might not be exact; its true angle can typically be within 10° of the angle of detection, even if that falls outside of the detection angle range. If M_FIT_MODE is disabled, the returned angle of an occurrence will always be the angle at which it was detected and will therefore always be within the detection angle range. If M_FIT_MODE is enabled, a fit operation is performed starting from the angle of detection. In refining the angle so that it is more accurate, the angle of the occurrence's reference axis returned by MagmGetResult() with M_ANGLE can end up being outside of the detection angle range. Note that the greater the range of angles to be searched, the slower the search. To set the upper limit for the detection angle range, use M_DETECTION_DELTA_POS_ANGLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the lower limit of the detection angle range, in degrees. Note that M_DETECTION_DELTA_NEG_ANGLE + M_DETECTION_DELTA_POS_ANGLE cannot be greater than 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DETECTION_DELTA_POS_ANGLE	Sets the upper limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *UpperLimit* = M_DETECTION_REF_ANGLE + M_DETECTION_DELTA_POS_ANGLE. The detection angle range should be used to cover the expected variance in the angle of detection. Note that the detection angle range only limits the angle of detection. The angle at which an occurrence is detected might not be exact; its true angle can typically be within 10° of the angle of detection, even if that falls outside of the detection angle range. If M_FIT_MODE is disabled, the returned angle of an occurrence will always be the angle at which it was detected and will therefore always be within the detection angle range. If M_FIT_MODE is enabled, a fit operation is performed starting from the angle of detection. In refining the angle so that it is more accurate, the angle of the occurrence's reference axis returned by MagmGetResult() with M_ANGLE can end up being outside of the detection angle range. Note that the greater the range of angles to be searched, the slower the search. To set the lower limit for the detection angle range, use M_DETECTION_DELTA_NEG_ANGLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the upper limit of the detection angle range, in degrees. Note that M_DETECTION_DELTA_NEG_ANGLE + M_DETECTION_DELTA_POS_ANGLE cannot be greater than 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b

M_DETECTION_MULTI_ANGLE_MODE		Sets the mode with which to detect occurrences when searching at multiple angles. The M_GLOBAL mode can accelerate search speed, but can also increase the risk of missing overlapping occurrences that have close centers. You should specify M_INDIVIDUAL if the search is not finding all model occurrences, but note that this will slow down the search. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_GLOBAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GLOBAL	Specifies to detect occurrences across all search angles globally.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INDIVIDUAL	Specifies to detect occurrences at each search angle individually.	c		e	f	g	a		i		j	k	l	m	n	b
M_DETECTION_REF_ANGLE		Sets the nominal search angle; this is the angle at which you expect to find the model's reference axis in the target. Note that occurrences can be slightly rotated from the search angle and still be detected. The exact degree varies depending on your model, but the MagmFind() operation will typically detect occurrences within 10° of the search angle. If you expect greater degrees in variation, you should specify a detection angle range, using M_DETECTION_DELTA_NEG_ANGLE and M_DETECTION_DELTA_POS_ANGLE. Note that specifying a nominal search angle does not decrease the search speed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* < 360.0	Specifies the nominal search angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_DETECTION_STEP_ANGLE		Sets the step angle at which to search for occurrences of the model within the specified detection angle range. Note that the smaller the step angle, the slower the search. The search is performed at the nominal search angle and at every +/- M_DETECTION_STEP_ANGLE degrees within the detection angle range. Note that occurrences can be slightly rotated from the search angle and still be detected. You should typically set the step angle to this rotation * 2. For example, if your model can tolerate the occurrence being offset from its search angle by ±5°, you should set the step angle to 10°. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
	5.0 <= *Value* <= 180.0	Specifies the step angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_EDGEL_ANGLE_MODE		Sets whether to consider edgel orientation, defined by a gradient angle, when searching for occurrences. A smaller difference in orientation between model and occurrence edgels produces a closer match. The gradient angle is the direction of maximum change in brightness at the edgel's position. Note that edgel orientation is indifferent to whether the brightness changes from dark to light or the reverse. This means that the maximum difference in orientation between two edgels is 90 degrees. Disabling M_EDGEL_ANGLE_MODE will increase the search speed. You can typically disable this control when your target images are uncluttered with minimal noise. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ORIENTATION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to consider edgel orientation.	c		e	f	g	a		i		j	k	l	m	n	b

		M_ORIENTATION	Specifies to consider edgel orientation. The difference in orientation between model and occurrence edgels is negatively correlated with the occurrence's detection, fit, and coverage scores. Smaller comparative differences result in higher scores (and vice versa).	c		e	f	g	a		i		j	k	l	m	n	b
	M_EMPTY_REGIONS_MODE		Sets whether to consider the regions of the model that do not contain edges when searching for model occurrences. If the model has a consistent region without edges, you can enable this control type to improve the detection score of favorable occurrences. If edges are present in this region of the occurrence, the detection score will decrease. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically determine the empty regions. If no empty regions are found, the outcome is the same as disabling empty regions.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to consider empty regions.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIRST_LEVEL	Sets the resolution level for the initial stage (lowest resolution) of the search when M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED. Level 0 is the original target and each higher level is half the size (and resolution) of the previous one. Note that very small models and search regions cannot be subsampled as many times as large ones. If the specified level results in a model that is too small or does not have enough edgels, an error will occur when calling MagmPreprocess(). A higher first level speeds up the initial search but might make it less reliable because the model might not retain enough distinctive features at a lower resolution. M_FIRST_LEVEL is for advanced users of the AGM module. The default setting usually provides the best results for the search operation. Note, M_FIRST_LEVEL cannot be lower than M_LAST_LEVEL. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 7	Specifies the resolution level. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_MODE	Sets the mode of the fit operation. The fit mode affects positional accuracy and search speed. The positional results are most accurate when the fit operation uses subpixel precision, but this can reduce the speed of the search. Note that the fit can only be calculated with subpixel precision when the edges of the target images are extracted (M_USE_MAGNITUDE_TARGET is disabled). You can disable the fit operation to speed up the search, in which case Aurora Imaging Library uses the features of the model and target to find occurrences, but it doesn't do any fitting. This results in a lower accuracy because the positional results are simply the positions of detection. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_PIXEL_PRECISION.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to disable the fit operation; the positional results are determined by detection only.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PIXEL_PRECISION	Specifies to calculate the fit with pixel precision.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SUBPIXEL_PRECISION	Specifies to calculate the fit with subpixel precision. This value is only available when M_USE_MAGNITUDE_TARGET is set to M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LAST_LEVEL		Sets the resolution level for the final stage (the highest resolution) of the search when M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED. Level 0 is the original target and each higher level is half the size (and resolution) of the previous one. Note that very small models and search regions cannot be subsampled as many times as large ones. If the specified level results in a model that is too small or does not have enough edgels, an error will occur when calling MagmPreprocess(). A higher last level speeds up the initial search but might make it less reliable and less accurate because the model might not retain enough distinctive features at a low resolution. M_LAST_LEVEL is for advanced users of the AGM module. The default setting usually provides the best results for the search operation. Note, M_LAST_LEVEL cannot be higher than M_FIRST_LEVEL. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <= 7	Specifies the resolution level. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MAX_ACCEPTED_OVERLAP	Sets the maximum allowable overlap between found occurrences, as a percentage. Note that M_MAX_ACCEPTED_OVERLAP is applied after possible occurrences are sorted. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the acceptable overlap, as a percentage. If set to 100%, found occurrences can overlap completely. If set to 0%, no found occurrence can overlap another by any amount.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX_ASSOCIATION_DISTANCE	Sets the maximum distance between a model edgel and the nearest occurrence edgel. Model edgels falling outside the specified maximum distance are not used to perform the occurrence's regular fit operation, but will decrease the overall fit score and the coverage score. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 4.0	Specifies the maximum association distance, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MODEL_PRECISION	Sets the percentage of model edgels to use when searching for occurrences. Specifying a high value can increase the detection, fit, and coverage scores, which increases the likelihood of finding all occurrences, but can also decrease the search speed. Specifying a low value can decrease the detection, fit, and coverage scores, which increases the likelihood of missing favorable occurrences or finding false occurrences, but can also increase the search speed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 80.0%.	c		e	f	g	a		i		j	k	l	m	n	b

		0.0 <= *Value* <= 100.0	Specifies the amount of model edgels, as a percentage. A value of 100% indicates that all model edgels are used.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERSEVERANCE_DETECTION		Sets the algorithm's perseverance when searching for occurrences. This affects the number of candidates the algorithm attempts to validate. Increasing the perseverance will increase robustness, but will increase search time. You should try increasing the perseverance if the search is not finding all model occurrences, or if the target is complex and decreasing M_STEP_X and M_STEP_Y is insufficient. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 90.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL_SCALE		Sets the pixel scale of the model, if the model was defined from a CAD DXF file. This value is the scale to apply to the model to go from DXF units to pixel units. For example, if M_BOX_SIZE_X is 50.0 DXF units and M_PIXEL_SCALE is set to 100.0, the model box will be 5000.0 pixels wide. This control type is only available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 < *Value* < 10000.0	Specifies the pixel scale. M_BOX_SIZE_... * M_PIXEL_SCALE must be less than or equal to 32768 pixels and M_EDGES_BOX_SIZE_... * M_PIXEL_SCALE must be between 6 and 32768 pixels to avoid an Aurora Imaging Library error when calling MagmPreprocess() or MagmDraw().	c		e	f	g	a		i		j	k	l	m	n	b

M_POLARITY	Sets the expected polarity of occurrences, compared to that of the model. This control type is not available for models of type M_DXF_FILE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANY	Specifies that the occurrences can be any polarity. That is, they can be the same or reverse of that of the model, or a mixture of polarities.	c		e	f	g	a		i		j	k	l	m	n	b
M_REVERSE	Specifies that the polarity of occurrences is the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies that the polarity of occurrences is the same as that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_PYRAMID_LEVELS_MODE	Sets how to establish the resolution levels for the search. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_AUTO_CONTENT_BASED.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO_CONTENT_BASED	Specifies to automatically determine the levels according to the model content.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the values set with M_FIRST_LEVEL and M_LAST_LEVEL.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_X	Sets the X-coordinate of the origin of the model's reference axis, relative to the model origin. Position results return the X-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies that the default X-coordinate will be used. For a model of type M_DXF_FILE, the default value is the origin of the CAD DXF file. For other types of models, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-coordinate, in pixels. This value can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Y		Sets the Y-coordinate of the origin of the model's reference axis, relative to the model origin. Position results return the Y-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the default Y-coordinate will be used. For a model of type M_DXF_FILE, the default value is the origin of the CAD DXF file. For other types of models, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-coordinate, in pixels. This value can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS		Sets the degree of smoothness (noise reduction) applied to the model image and target image(s) during the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. A setting of 0.0 indicates almost no noise reduction effect, while a setting of 100.0 indicates a very strong noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b

M_SORT_CANDIDATES_SCORE		Sets the score used to sort candidates. Candidates are sorted in decreasing score order. The specified score affects which possible occurrences are filtered out. When two possible occurrences overlap above the value specified with M_MAX_ACCEPTED_OVERLAP, the possible occurrence with the lower score is removed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SCORE_COVERAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_COVERAGE	Specifies to use the coverage score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_DETECTION	Specifies to use the detection score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT	Specifies to use the fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT_OVERALL	Specifies to use the overall fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
M_STABLE_EDGELS_MODE		Sets whether to use only stable edgels when searching for occurrences. A stable edgel is an edgel that remains persistent when extracting edges with slightly different smoothness values. This control type is only available for models of type M_IMAGE and M_IMAGE_REGION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SMOOTHNESS_STABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to use only stable edgels. All edgels are considered when searching for occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS_STABLE	Specifies to use only stable edgels. Unstable (noisy) edgels are ignored when searching for occurrences.	c		e	f	g	a		i		j	k	l	m	n	b

M_STEP_X		Sets the X step size of explored positions to search for occurrences. Increasing M_STEP_X and M_STEP_Y can accelerate search speed, but can also increase the risk of missing favorable occurrences. You can typically start with a value of 1 and increase M_STEP_X and M_STEP_Y if the speed is too slow. Increasing these values when you have a large model can decrease the search time while minimally affecting the occurrences' detection, fit, and coverage scores. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the X step size.	c		e	f	g	a		i		j	k	l	m	n	b
M_STEP_Y		Sets the Y step size of explored positions to search for occurrences. Increasing M_STEP_X and M_STEP_Y can accelerate search speed, but can also increase the risk of missing favorable occurrences. You can typically start with a value of 1 and increase M_STEP_X and M_STEP_Y if the speed is too slow. Increasing these values when you have a large model can decrease the search time while minimally affecting the occurrences' detection, fit, and coverage scores. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the Y step size.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_MODE		Sets the threshold mode of the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies no threshold. All edges are extracted.	c		e	f	g	a		i		j	k	l	m	n	b

		M_HIGH	Specifies a high threshold. M_HIGH should be used for images with some contrast variations, noise, and non-uniform illumination. M_HIGH always results in a lower (or equal) number of edgels than M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. All edges over a minimum noise-based estimated threshold are extracted. M_LOW always results in a lower (or equal) number of edgels than M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. M_MEDIUM should be used for multi-contrast images, or for images with a lot of noise or non-uniform illumination. M_MEDIUM always results in a lower (or equal) number of edgels than M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high threshold. Only the strongest edges in the image are extracted. M_VERY_HIGH always results in a lower (or equal) number of edgels than M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b

	M_USE_MAGNITUDE_TARGET	Sets whether to use the gradient magnitude or to use extracted edges when searching for occurrences. When this control type is set to M_ENABLE, thresholding is not applied to the target images, and the magnitude of the gradient is used instead. This is useful when there are contrast variations within or between images, such that there is no single threshold capable of detecting all pertinent edgels without introducing noisy background edges, leading to false positives. When set to M_DISABLE, edges are extracted from the target images based on a thresholding of the magnitude values. When the occurrences have strong contrasted edges that can be detected without introducing noisy background edges, using edges can lead to more robust results. Note that even if this control type is enabled, M_THRESHOLD_MODE is used to extract the edges of the model. This control type is not available when M_FIT_MODE is set to M_SUBPIXEL_PRECISION or when using an Edge Finder result buffer as the target. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to extract and use edges.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use the magnitude of the gradient.	c		e	f	g	a		i		j	k	l	m	n	b
Train AGM context ID with a composite-definition model			Specifies a train AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_TRAIN and used in MagmTrain() operations. It must contain a composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b

M_EMPTY_REGIONS_MODE	Sets whether to consider the regions of the model appearances that do not contain edges when training the composite-definition model. This will slow down training, but if the model has a consistent region without edges, using empty regions can improve the detection score of favorable occurrences. If edges are present in this region of the occurrence, the detection score will decrease. The setting of this control type is also used for the MagmFind() operation when searching for the trained composite-definition model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically determine the empty regions. If no empty regions are found, the outcome is the same as disabling empty regions.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to consider empty regions.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_PRECISION	Sets the percentage of the trained composite-definition model's edgels to use when searching for occurrences. Specifying a high value can increase the detection, fit, and coverage scores, which increases the likelihood of finding all occurrences, but can also decrease the search speed. Specifying a low value can decrease the detection, fit, and coverage scores, which increases the likelihood of missing favorable occurrences or finding false occurrences, but can also increase the search speed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 80.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the amount of model edgels, as a percentage. A value of 100% indicates that all model edgels are used.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NEGATIVE_LABELS_MAX_OVERLAP	Sets the maximum allowable overlap between two automatically generated negative (red) rectangles. This control type is only used when M_NEGATIVE_LABELS_MODE is set to M_AUTO. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* < 90.0	Specifies the maximum allowable overlap, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE_LABELS_MODE	Sets whether to automatically label negative model locations in the training images or to use manually defined labels. If set to M_AUTO, MagmTrain() will automatically identify and label background areas with red rectangles. Note that the number of automatically generated red rectangles depends on the complexity of the training images. You can use MagmGetResult() with M_NUMBER_NEG_RECTANGLES to retrieve the number of red rectangles across all training images. You can use MagmDraw() to view the rectangles. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ACCORDING_TO_LABELS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCORDING_TO_LABELS	Specifies to use manually defined negative (red) rectangles if there is at least one red rectangle in the set of training images; otherwise, specifies to automatically label the negative model locations.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies to automatically label the negative model locations. At least one training image must have a minimum of one blue rectangle, and no training images can have any red rectangles. Note, there can be training images with no rectangles.	c		e	f	g	a		i		j	k	l	m	n	b

		M_USER_DEFINED	Specifies to use manually defined negative (red) rectangles. At least one training image must have a minimum of one red rectangle, and at least one training image must have a minimum of one blue rectangle. Note, there can be training images with only red rectangles or only blue rectangles. You can also include training images with no rectangles, but they will not be used during training.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE_LABELS_POSITIVE_MAX_OVERLAP		Sets the maximum allowable overlap between an automatically generated negative (red) rectangle and a positive (blue) rectangle. This control type is only used when M_NEGATIVE_LABELS_MODE is set to M_AUTO. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.0%.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* < 80.0	Specifies the maximum allowable overlap, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REALIGN_STRENGTH		Sets the strength with which to displace the blue rectangles, which define the positive model locations in the training images, to better align them. The more aligned the blue rectangles are, the more accurate the search with the resulting trained composite-definition model will be. You should attempt to label your training images such that all model occurrences appear in the exact same position within each blue rectangle. If the blue rectangles are not precise, you can use this control to allow the MagmTrain() operation to displace the blue rectangles. The higher the realignment strength, the further MagmTrain() can displace the blue rectangles so that the model occurrences appear in the same position within each blue rectangle. When the realignment strength is set to 0.0, no realignment occurs. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b

		0.0 <= *Value* <= 100.0	Specifies the strength of the realignment.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS		Sets the degree of smoothness (noise reduction) applied to the training images during the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. A setting of 0.0 indicates almost no noise reduction effect, while a setting of 100.0 indicates a very strong noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE		Sets the threshold mode of the edge extraction. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies no threshold. All edges are extracted.	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies a high threshold. M_HIGH should be used for images with some contrast variations, noise, and non-uniform illumination. M_HIGH always results in a lower (or equal) number of edgels than M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. All edges over a minimum noise-based estimated threshold are extracted. M_LOW always results in a lower (or equal) number of edgels than M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. M_MEDIUM should be used for multi-contrast images, or for images with a lot of noise or non-uniform illumination. M_MEDIUM always results in a lower (or equal) number of edgels than M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b

		M_VERY_HIGH	Specifies a very high threshold. Only the strongest edges in the image are extracted. M_VERY_HIGH always results in a lower (or equal) number of edgels than M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRAIN_DETECTION_ANGLE		Sets whether to train the composite-definition model using rotated labeled image regions (red and blue rectangles). You should enable this control to increase the likelihood of finding occurrences at any angle. A composite-definition model trained without rotation is more likely to miss rotated occurrences. Note that if this control type is set to M_DISABLE, calling MagmTrain() generates an error if your training images contain rotated labeled image regions. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_ACCORDING_TO_LABELS.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ACCORDING_TO_LABELS	Specifies to use rotated labeled image regions if there is at least one rotated labeled image region in the set of training images; otherwise, specifies not to use rotated labeled image regions.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to use rotated labeled image regions. All labeled image regions must have an angle of 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use rotated labeled image regions. Labeled image regions can have any angle.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRAIN_LABELS_NOISE_MODE	Sets the mode with which to handle substandard labeled image regions when training the composite-definition model. If this control type is set to M_LABELS_CONFIDENCE, the MagmTrain() operation will be robust to substandard labeled image regions that might cause distortion due to the presence of a lot of noise, blue rectangles (positive model locations) that have missing or extra features in comparison to other blue rectangles, red rectangles (negative model locations) that contain similar features to the model, or incorrect labeling (model appearances labeled with negative (red) rectangles or background samples labeled with positive (blue) rectangles). Disabling M_TRAIN_LABELS_NOISE_MODE can increase the training speed, but can result in a more distorted trained composite-definition model if the training images contain any substandard labeled image regions, as listed above. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_LABELS_CONFIDENCE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to handle substandard labeled image regions differently. MagmTrain() will strongly attempt to correctly classify all labeled regions, including outlier labeled regions, incorrect labeled regions, or labeled regions with a lot of noise, which can cause the model to overfit on the training data.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LABELS_CONFIDENCE	Specifies to be robust to substandard labels. To prevent overfitting on the training data, labels that are hard to correctly classify, such as outlier labels, incorrect labels, or labels with a lot of noise, will have less of an affect on the resulting trained composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b

	M_USE_MAGNITUDE_TARGET	Sets whether to use the gradient magnitude or to use extracted edges when training the composite-definition model and searching for occurrences. When this control type is set to M_ENABLE, thresholding is not applied to the training images or target images, and the magnitude of the gradient is used instead. This is useful when there are contrast variations within or between images, such that there is no single threshold capable of detecting all pertinent edgels without introducing noisy background edges, leading to false positives. When set to M_DISABLE, edges are extracted from the training and target images based on a thresholding of the magnitude values. When the occurrences have strong contrasted edges that can be detected without introducing noisy background edges, using edges can lead to more robust results. Note that even if this control type is enabled, M_THRESHOLD_MODE will be copied to the find AGM context, and used to extract the edges of the model when M_MODEL_SOURCE is set to M_USER_IMAGE. This control type is not available when M_FIT_MODE is set to M_SUBPIXEL_PRECISION or when using an Edge Finder result buffer as the target. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to extract and use edges.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use the magnitude of the gradient.	c		e	f	g	a		i		j	k	l	m	n	b

The following ContextOrResultId, ControlType, and ControlValue parameter settings can be specified for a train AGM result buffer. In this case, you must set the Index parameter to M_GENERAL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a train AGM result buffer																			
ContextOrResultId - Possible values returned			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlType																		
		ControlValue																	

Train AGM result buffer ID			Specifies a train AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT.														
	M_STOP_TRAIN		Stops the current execution of MagmTrain(). This can only be done from another thread of higher priority. Results already available in the result buffer become invalid and will be discarded.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MagmControlInt64 (AIL_ID ContextOrResultId, AIL_INT64 Index, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MagmControlDouble (AIL_ID ContextOrResultId, AIL_INT64 Index, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmCopyResult

Synopsis

Copy a trained composite-definition model from a train AGM result buffer to a find AGM context.

Syntax

```
void MagmCopyResult(
    AIL_ID SrcTrainResultId, //in
    AIL_INT64 SrcIndex,      //in
    AIL_ID DstContextId,     //out
    AIL_INT64 DstIndex,      //in
    AIL_INT64 CopyType,      //in
    AIL_INT64 ControlFlag    //in
)
```

Description

This function copies a model from a train AGM result buffer to a find AGM context. You can only copy trained results after calling MagmTrain().

You must use **MagmCopyResult**() to add a composite-definition model to a find AGM context. Once added, you must preprocess the context before calling MagmFind().

Parameters

SrcTrainResultId

Specifies the identifier of the source result buffer from which to copy. The result buffer must have been allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT, and must contain the results of a call to MagmTrain().

SrcIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

DstContextId

Specifies the identifier of the destination context in which to store the copied results. The context must have been allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND.

DstIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

CopyType

Specifies the type of copy operation to perform.

Specifies the type of copy operation	
Value	Description
M_TRAINED_MODEL	Specifies to copy the trained composite-definition model from the train AGM result buffer to the find AGM context.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmDefine

Synopsis

Add a model to, or delete a model from, an AGM context.

Syntax

```
AIL_INT64 MagmDefine(
    AIL_ID ContextId,     //out
    AIL_INT64 Operation,  //in
    AIL_INT64 SourceType, //in
    AIL_INT64 ModelType,  //in
    AIL_INT64 Param1,     //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function allows you to add a model to, or delete a model from, a find AGM context or a train AGM context. Note, when a model is added or deleted, the AGM context must be preprocessed again, using MagmPreprocess(). You can only define one model per AGM context. When adding a model to the context, it is assigned an index of 0.

Note that a composite-definition model must be trained before performing an MagmFind() operation. This means you cannot add a composite-definition model to a find AGM context using **MagmDefine**(). You must add it to a train AGM context using **MagmDefine**(), train the model using MagmTrain(), and then copy the trained model to a find AGM context using MagmCopyResult().

The MagmFind() and MagmTrain() operations are performed according to the individual model settings specified using MagmControl().

Parameters

ContextId

Specifies the identifier of the AGM context in which to add or delete a model. The AGM context must have been previously allocated on the required system using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND or M_GLOBAL_EDGE_BASED_TRAIN.

Operation

Specifies whether to add or remove a model.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies the operation																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ADD	Specifies to add a model to the specified AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DELETE	Specifies to delete a model from the specified AGM context.	c		e	f	g	a		i		j	k	l	m	n	b

SourceType

Specifies the type of source from which to define the model.

See the Parameter associations section for possible values that can be specified.

ModelType

Specifies the type of model to define when adding models to the AGM context.

See the Parameter associations section for possible values that can be specified.

Param1

Specifies an attribute of the model.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The tables below list possible values for the **SourceType**, **ModelType**, and **Param1** parameters.

- For adding a model to the AGM context
- For removing a model from the AGM context

To add a model to an AGM context, the ModelType, SourceType, and Param1 parameters can be set to the following values. The Operation parameter must be set to M_ADD.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For adding a model to the AGM context																	
ModelType			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	SourceType																	
		Param1																
M_COMPOSITE			Specifies to add a composite-definition model. A composite-definition model uses training images, labeled with model appearances and background areas that can be incorrectly misconstrued as occurrences, to construct the best possible model. A composite-definition model can be used to improve results if a previous MagmFind() operation with a single-definition model found false occurrences. Training a composite-definition model, using MagmTrain(), identifies possible backgrounds, which can result in a model that can better discriminate true occurrences from false occurrences, especially in cluttered target images. Since a composite-definition model requires training before it can be used to search for occurrences, this type of model can only be added to a train AGM context.															

M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that this parameter is unused. You do not specify a model image because the model is built from training images, using MagmTrain().	c		e	f	g	a		i		j	k	l	m	n	b
M_SINGLE	Specifies to add a single-definition model. A single-definition model can be defined from a CAD DXF file, Edge Finder result buffer, or a single source image. In the latter case, the entire image is used as the model or the model is extracted from a rotated or non-rotated region. Since a single-definition model does not require training, this type of model can only be added to a find AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DXF_FILE	Specifies to define the model from the entities in the specified CAD DXF file. This function does not support all entities that are possible in a CAD DXF file. If there are unsupported entities in the file, they are ignored, and the rest of the entities are read. The supported entities are as follows: LINE, ARC, CIRCLE, ELLIPSE, INSERT, POLYLINE, and LWPOLYLINE (including bulge information). For INSERT entities, only single insertions are supported; multiple insertions using the MINSERT command are not supported. Extrusion is also not supported. A model defined from a CAD DXF file has no polarity.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*DxfFilePath*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*", for example: "*remote:///C:\mydirectory\myfile*". > Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*DxfFilePath*")).	c		e	f	g	a		i		j	k	l	m	n	b

	M_EDGE_RESULT	Specifies to define the model from the results of an edge extraction performed using Edge Finder. To define a model from an Edge Finder result buffer, the buffer must have been previously allocated using MedgeAllocResult(). MedgeCalculate() must have been already called. In addition, the Edge Finder result buffer must be compatible with the AGM context. Enable compatibility using MedgeControl() with M_AGM_COMPATIBLE prior to calling MedgeCalculate(). The model is defined to have all currently included edges in the result buffer. In the case of an Edge Finder-type model, the model image is the image from which the Edge Finder results were obtained.	c		e	f	g	a		i		j	k	l	m	n	b
		Edge Finder result buffer identifier	Specifies the identifier of the Edge Finder result buffer. The Edge Finder result buffer must be allocated on the same system as the AGM context. If it is not, an error will occur. Note that because AGM does not support calibration, the Edge Finder result buffer must be uncalibrated or M_RESULT_OUTPUT_UNITS must be set to M_PIXEL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_IMAGE	Specifies to define the model from the specified model source image. The entire image is used as the model. To define the model from a region of the model source image, use M_IMAGE_REGION instead.	c		e	f	g	a		i		j	k	l	m	n	b
		Image buffer identifier	Specifies the identifier of the image buffer containing the model image. The image buffer must be a 1-band 8-bit unsigned buffer. The minimum and maximum model image sizes are 6x6 and 32768x32768 pixels, respectively. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error. Note that because AGM does not support calibration, the image buffer must be uncalibrated.	c		e	f	g	a		i		j	k	l	m	n	b

M_IMAGE_REGION	Specifies to define the model from a region of the specified model source image. Only edgels inside the region will be part of the model. Note that the entire model source image contributes to determining the numerical threshold value used to extract edges from the region. To extract the model from a rotated region in the model source image, pass a model source image that is associated to a rectangular region of interest (ROI) at the required angle.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the identifier of the image buffer from which to extract the model. The image buffer must be a 1-band 8-bit unsigned buffer and must have a (rotated or non-rotated) rectangular region of interest (ROI) associated with it, defined using MbufSetRegion() with a 2D graphics list and either M_RASTERIZE or M_NO_RASTERIZE. An error will be generated if the ROI is only in raster format (M_RASTER) or has a non-rectangular shape. The minimum and maximum sizes are 6x6 and 32768x32768 pixels, respectively, for both the model source image and the ROI. Note that the model's size is determined by the size of the specified region. The center of the region must be inside the model source image. Note that because AGM does not support calibration, the image buffer must be uncalibrated.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

To delete a model from an AGM context, the ModelType, SourceType, and Param1 parameters must be set to the following values. The Operation parameter must be set to M_DELETE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For removing a model from the AGM context																
ModelType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
SourceType																
Param1																
M_DEFAULT	Specifies the default behavior.															
M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_MODEL_INDEX(AIL_INT Index)	Specifies the model in the AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

				Index Set this parameter to the following:																	
					0	Specifies the index of the model to delete.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

Reserved for future expansion and returns **M_NULL**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmDraw

Synopsis

Draw specific features of the model, result occurrence, or training image in an image buffer or 2D graphics list.

Syntax

```
void MagmDraw(
    AIL_ID ContextGraId,           //in
    AIL_ID ContextOrResultAgmId,   //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_INT64 Operation,           //in
    AIL_INT64 Index,               //in
    AIL_INT64 ControlFlag          //in
)
```

Description

This function draws specific model, result occurrence, or training image features in the destination image buffer or 2D graphics list.

Drawing the trained composite-definition model will help you determine the success of the MagmTrain() operation. If the trained composite-definition model does not look like the positively labeled objects in your training images, you should modify the images and repeat the training process.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context to use when drawing. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextOrResultAgmId

Specifies the find AGM context or AGM result buffer from which to extract the features to draw. The find AGM context must have been previously allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND. The find or train AGM result buffer must have been previously allocated using MagmAllocResult() with

M_GLOBAL_EDGE_BASED_FIND_RESULT or M_GLOBAL_EDGE_BASED_TRAIN_RESULT and must contain the results of a call to MagmFind() or MagmTrain(), respectively.

DstImageBufOrListGraId

Specifies the identifier of the destination image buffer or 2D graphics list in which to draw. The buffer can be any valid 1- or 3-band image buffer allocated using MbufAlloc...(). The 2D graphics list must be previously allocated using MgraAllocList(). By drawing into a display's overlay buffer or associating the 2D graphics list with the display, you can also annotate an image non-destructively.

If specifying an image buffer, it must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

When drawing a model, the destination image must be at least the size of the model. Use MagmInquire() with M_SIZE_X and M_SIZE_Y to get the correct size.

Operation

Specifies the type of operation to perform. Operations can be added together to draw multiple features at a time. For example, to draw both the result occurrence's position and edges, you would set the Operation parameter to M_DRAW_POSITION + M_DRAW_EDGES. The Operation parameter can be set to a combination of the following values:

The following operations can extract the features to draw only from a find AGM context and cannot draw in a 2D graphics list.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a find AGM context only																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_IMAGE +	Draws the model image of a single-definition or a composite-definition model. For a composite-definition model, if the image does not exist, an Aurora Imaging Library error is thrown. To inquire if the model image exists, you can use MagmInquire() with M_HAS_USER_MODEL. You can use MagmInquire() with M_SIZE_X and M_SIZE_Y to inquire the size of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TRAINED_MODEL	Draws the trained composite-definition model. You can use MagmInquire() with M_SIZE_X and M_SIZE_Y to inquire the size of the model.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values that can be used alone or as combination values for M_DRAW_IMAGE.

You can use one or more of the following values in combination with each other or with the above-mentioned value, to specify an operation that can extract features to draw from either a find AGM context or a find AGM result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying an operation that can extract features to draw from either a find AGM context or a find AGM result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DRAW_BOX	Draws the model box, or a bounding box around the occurrence. Note that when drawing an occurrence, the bounding box is drawn according to the occurrence's angle and scale. Since a find AGM result buffer only contains occurrences that are similar in size to the model, the occurrence's scale is close to 1.0. You can get the exact angle and scale of the occurrence, using MagmGetResult() with M_ANGLE and M_SCALE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EDGES +	Draws the edges related to the model or result. When working with result occurrences, the drawing depends on whether you are adding M_MODEL or M_TARGET to M_DRAW_EDGES. Note that the AGM context must be preprocessed. If it is not preprocessed, an Aurora Imaging Library error is thrown.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_POSITION	Draws a cross-like symbol at the model's reference axis origin or occurrence position. The cross is drawn maintaining the angle of the model's reference axis or the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_DRAW_EDGES.

You can add one of the following values to the above-mentioned value to specify whether to draw the active edges of the model transformed at the occurrence position or to draw the edges of the target in the region of the occurrence.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For M_DRAW_EDGES when working with result occurrences																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MODEL	Draws the used edges of the model transformed at the occurrence position. For a composite-definition model, if MagmControl() with M_MODEL_SOURCE is set to M_USER_IMAGE, the model image is used for the fit and coverage calculations and the edges of the model image are drawn. If M_MODEL_SOURCE is set to M_FROM_TRAIN, the trained composite-definition model is used for the detection, fit, and coverage calculations and the edges of the trained composite-definition model are drawn. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET	Draws the edges of the target in the region of the occurrence. This value is only available when M_USE_MAGNITUDE_TARGET is set to M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

The following operations can extract the features to draw from a train AGM result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro

Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a train AGM result buffer only																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_NEG_RECTANGLES	Draws all the rectangles that were used to label the negative model locations in the training image. You can use MgraControl() with M_COLOR set to M_COLOR_RED to reuse these rectangles for another MagmTrain() operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_POS_RECTANGLES	Draws all the rectangles that were used to label the positive model locations in the training image. Note that if the rectangles were realigned, using MagmControl() with M_REALIGN_STRENGTH, the realigned rectangles are drawn. You can use MgraControl() with M_COLOR set to M_COLOR_BLUE to reuse these rectangles for another MagmTrain() operation.	c		e	f	g	a		i		j	k	l	m	n	b

Index

Specifies the index of the model in the specified find AGM context, the result occurrence in the specified find AGM result buffer, or the image in the specified train AGM result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro
Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the model, result occurrence, or image																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL. Note that this value is only available if a find AGM result buffer is specified.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_IMAGE_INDEX(AIL_INT Index)	Specifies that the drawing operation is for a specific image in the train AGM result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Set this parameter to the following:															

			Value >= 0 Specifies the index of the training image. This value must be less than the value passed to the NumImages parameter of MagmTrain().	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_MODEL_INDEX(AIL_INT Index)	Specifies that the drawing operation is for a specific model in the find AGM context, if one is specified.			c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Index Set this parameter to the following:															
			0 Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Draws the specified feature of all occurrences, if a find AGM result buffer is specified.			c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the index of the occurrence in the find AGM result buffer, if one is specified.			c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. this parameter must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmFind

Synopsis

Find the specified model in the target image or Edge Finder result buffer.

Syntax

```
void MagmFind(
    AIL_ID ContextAgmId,          //in
    AIL_ID TargetImageOrResultId, //in
    AIL_ID ResultAgmId,           //out
    AIL_INT64 ControlFlag         //in
)
```

Description

This function searches for the model, defined in the specified find AGM context, in the specified target image or Edge Finder result buffer. Results are stored in the specified find AGM result buffer. You can retrieve results using MagmGetResult(). Note that calling **MagmFind**() successively with target images of different sizes will slow the execution time.

This function only searches for single-definition models or trained composite-definition models. You can train a composite-definition model using MagmTrain(), and then copy the resulting model to a find AGM context using MagmCopyResult() with M_TRAINED_MODEL.

Use MagmControl() to configure individual model settings before calling this function.

An occurrence is selected as a candidate if its coverage, fit, and detection scores are greater than or equal to the respective acceptance levels (MagmControl() with M_ACCEPTANCE_COVERAGE, M_ACCEPTANCE_FIT, and M_ACCEPTANCE_DETECTION).

This function returns results in decreasing coverage, fit, or detection score order, depending on MagmControl() with M_SORT_CANDIDATES_SCORE. This means that the most-likely occurrence is always returned first.

If you are searching for the model in an Edge Finder result buffer, **MagmFind**() will search for the model only in the included edges of the Edge Finder result buffer.

Note that AGM does not provide support for camera calibration. If you are searching for the model in a target image buffer, the target image buffer must be uncalibrated. If you are searching in an Edge Finder result buffer, the Edge Finder result buffer must be uncalibrated or M_RESULT_OUTPUT_UNITS must be set to M_PIXEL.

The AGM context must be preprocessed with MagmPreprocess() before calling this function.

Parameters

ContextAgmId

Specifies the find AGM context identifier to use for the search. The AGM context must have been previously allocated on the system using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND.

TargetImageOrResultId

Specifies the identifier of the target image or Edge Finder result buffer in which to find the model.

For specifying the target in which to search																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Edge Finder result buffer identifier	Specifies the identifier of the Aurora Imaging Library Edge Finder result buffer in which to find the model. The Edge Finder result buffer must have been previously allocated on the required system using MedgeAllocResult(); MedgeCalculate() must have already been called. In addition, the Edge Finder result buffer must be compatible with the AGM context. Enable compatibility using MedgeControl() with M_AGM_COMPATIBLE prior to calling MedgeCalculate(). Note that because AGM does not support calibration, the Edge Finder result buffer must be uncalibrated or M_RESULT_OUTPUT_UNITS must be set to M_PIXEL. This value is not available when M_USE_MAGNITUDE_TARGET is set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the identifier of the Aurora Imaging Library image buffer in which to find the model. The image buffer must be a 1-band 8-bit unsigned buffer. The minimum and maximum target image sizes are 6x6 and 32768x32768 pixels, respectively. It is important to note that the minimum and maximum target image sizes are Aurora Imaging Library limits. To make sure that you have enough memory, you can call **MagmFind**(); if you don't have enough memory, you will get an Aurora Imaging Library error. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error. Note that because AGM does not support calibration, the image buffer must be uncalibrated.	c		e	f	g	a		i		j	k	l	m	n	b

ResultAgmId

Specifies the identifier of the find AGM result buffer in which to store results. The result buffer must have been allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_FIND_RESULT.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmFree

Synopsis

Free an AGM context or result buffer.

Syntax

```
void MagmFree(
    AIL_ID ContextOrResultAgmId //in
)
```

Description

This function deletes the specified AGM context (and its model) or result buffer identifier, and releases any memory associated with it. Note that to only delete individual models in the context, use MagmDefine().

All AGM contexts and result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ContextOrResultAgmId

Specifies the identifier of the AGM context or result buffer to free. These must have been successfully allocated (with MagmAlloc() or MagmAllocResult()) prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmGetResult

Synopsis

Get results from a find or train AGM result buffer.

Syntax

```
AIL_DOUBLE MagmGetResult(
    AIL_ID ResultAgmId,   //in
    AIL_INT64 Index,      //in
    AIL_INT64 ResultType, //in
    void *ResultArrayPtr  //out
)
```

Description

This function retrieves the results of the specified type for the specified occurrence(s) from a find AGM result buffer, or for the specified model or specified rectangle from a train AGM result buffer. For find AGM result buffers, results are available after calling MagmFind(). For train AGM result buffers, results are available after calling MagmTrain().

Parameters

ResultAgmId

Specifies the identifier of the AGM result buffer from which to retrieve results. The result buffer must have been previously allocated on the system using MagmAllocResult() with M_GLOBAL_EDGE_BASED_FIND_RESULT or M_GLOBAL_EDGE_BASED_TRAIN_RESULT.

See the Parameter associations section for possible values that can be specified.

Index

Specifies where to get results.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying where to get results																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_MODEL_INDEX(AIL_INT ModelIndex)	Specifies to retrieve results for the model in the train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ModelIndex Set this parameter to the following:															

			0 Specifies the index of the model in the train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_RECTANGLE_NEG_INDEX(AIL_INT Index)	Specifies to retrieve results for a specific negative (red) rectangle in the train AGM result buffer.			c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Index Set this parameter to the following:															
			0 <= *Value* < M_NUMBER_NEG_RECTANGLES Specifies the index of the negative (red) rectangle in the train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_RECTANGLE_POS_INDEX(AIL_INT Index)	Specifies to retrieve results for a specific positive (blue) rectangle in the train AGM result buffer.			c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Index Set this parameter to the following:															
			0 <= *Value* < M_NUMBER_POS_RECTANGLES Specifies the index of the positive (blue) rectangle in the train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

M_ALL	Specifies to retrieve results for all model occurrences in the find AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve results relating to the entire find or train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *OccIndex* < M_NUMBER	Specifies the index of the occurrence in the find AGM result buffer for which to retrieve results.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address of the array in which to write results.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultAgmId** and **ResultType** parameters and possible values returned to the **ResultArrayPtr** parameter.

- For retrieving general results from a find or train AGM result buffer
- For retrieving a result of a single occurrence or all occurrences from a find AGM result buffer
- For retrieving a transformation coefficient from a find AGM result buffer
- For retrieving rectangle results from a train AGM result buffer
- For retrieving model results from a train AGM result buffer

To retrieve general results from a find or train AGM result buffer, set ResultType to one of the following values. In this case, set the Index parameter to M_GENERAL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving general results from a find or train AGM result buffer																	
ResultAgmId		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultType																

ResultArrayPtr - Possible values returned																
Find AGM result buffer ID for general results	Specifies a find AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_FIND_RESULT, and used to store MagmFind() results.															
M_NUMBER +	Retrieves the number of occurrences found.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS +	Retrieves the status of the find operation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CALCULATE_NOT_PERFORMED	Specifies that a find operation has not been performed. This is the initial status of a find AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETE	Specifies that the find operation completed.	c		e	f	g	a		i		j	k	l	m	n	b
M_CURRENTLY_CALCULATING	Specifies that the find operation is ongoing. You can only get this status if you are retrieving it from another thread.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERNAL_ERROR	Specifies that an unexpected error occurred.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_ENOUGH_MEMORY	Specifies that the operation was not completed because of a lack of available memory.	c		e	f	g	a		i		j	k	l	m	n	b
Train AGM result buffer ID for general results	Specifies a train AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT, and used to store MagmTrain() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +	Retrieves the number of trained composite-definition models. Note that training results in only 0 or 1 trained composite-definition models.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER_NEG_RECTANGLES +	Retrieves the number of negative (red) rectangles across all training images.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_POS_RECTANGLES +	Retrieves the number of positive (blue) rectangles across all training images.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS +	Retrieves the status of the train operation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETE	Specifies that the train operation completed. Note that an M_COMPLETE result from a train AGM result buffer does not mean that training attempts on the model were successful. To check the status of an individual model, set the Index parameter to the required model's index.	c		e	f	g	a		i		j	k	l	m	n	b
M_CURRENTLY_TRAINING	Specifies that the train operation is ongoing. You can only get this status if you are retrieving it from another thread.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERNAL_ERROR	Specifies that an unexpected error occurred.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_ENOUGH_MEMORY	Specifies that the operation was not completed because of a lack of available memory.	c		e	f	g	a		i		j	k	l	m	n	b
M_STOPPED_BY_REQUEST	Specifies that the train operation was stopped from another thread using MagmControl() with M_STOP_TRAIN.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRAINING_NOT_PERFORMED	Specifies that a train operation has not been performed. This is the initial status of a train AGM result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve results for a single occurrence or for all occurrences from a find AGM result buffer, set ResultType to one of the following values. In this case, set the Index parameter to M_ALL or the index of the occurrence.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving a result of a single occurrence or all occurrences from a find AGM result buffer																		
ResultAgmId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultType																
ResultArrayPtr - Possible values returned																
Find AGM result buffer ID for occurrence-specific results	Specifies a find AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_FIND_RESULT, and used to store MagmFind() results.															
M_ANGLE +	Retrieves the angle of the occurrence's reference axis, relative to the pixel coordinate system. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY +	Retrieves the polarity of the occurrence. Note that occurrences are only found if they match the polarity set using MagmControl() with M_POLARITY.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_ANY	Specifies that MagmControl() with M_POLARITY is set to M_ANY and the polarity of the occurrence is not determined. It could be the same or reverse of that of the model, a mixture of polarities, or no polarity in the case of an M_DXF_FILE model occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_REVERSE	Specifies that MagmControl() with M_POLARITY is set to M_REVERSE and the polarity of the occurrence is the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies that MagmControl() with M_POLARITY is set to M_SAME and the polarity of the occurrence is the same as that of the model.	c		e	f	g	a		i		j	k	l	m	n	b

	M_POSITION_X +	Retrieves the X-coordinate of the occurrence. This is the X-position of the model's reference axis transformed at the occurrence. Note, the origin of the model's reference axis is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_Y +	Retrieves the Y-coordinate of the occurrence. This is the Y-position of the model's reference axis transformed at the occurrence. Note, the origin of the model's reference axis is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCALE +	Retrieves the scale of the occurrence. Since the returned scale is uniform, it can be defined as the occurrence's size divided by the model's size. Note that only occurrences with small scale differences can be found, which means the returned scale will always be close to 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_COVERAGE +	Retrieves the coverage score of the occurrence, where the coverage score is the percentage of the total length of the model's edges found in the occurrence. 100% indicates that for each of the model's edges, a corresponding edge was found in the occurrence. Note, a model's edge corresponds to an edge in the occurrence if the distance between them is less than or equal to M_MAX_ASSOCIATION_DISTANCE.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_DETECTION +	Retrieves the detection score of the occurrence, where the detection score represents the likelihood that the found occurrence is a true occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT +		Retrieves the fit score of the occurrence, where the fit score is a measure of the correlation of the edges in the model to their corresponding edges in the occurrence. The score is calculated as follows: Fit Score = 1 - Normalized fit error The fit error is the average distance, in pixels, between the edges in the model and their corresponding edges in the occurrence. Note, only those model edges that have a corresponding edge in the occurrence are used in the fit score calculation. A model's edge has a corresponding edge in the occurrence if the distance between them is less than or equal to M_MAX_ASSOCIATION_DISTANCE.	c		e	f	g	a		i		j	k	l	m	n	b
		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SCORE_FIT_OVERALL +		Retrieves the overall fit score of the occurrence, where the overall fit score is a measure of the correlation of all edges in the model to those of the occurrence. The score is calculated as follows: Overall fit Score = 1 - Normalized overall fit error The overall fit error is the average distance, in pixels, between the edges in the model and edges in the occurrence. Note, all model edges are used in the overall fit score calculation, including those without a corresponding edge in the occurrence. For composite-definition models, this result type is only available when M_MODEL_SOURCE is set to M_USER_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve a transformation coefficient for a single occurrence or for all occurrences from a find AGM result buffer, set ResultType to one of the following values. In this case, set the Index parameter to M_ALL or the index of the occurrence.

These coefficients allow you to convert coordinates in the model coordinate system to the corresponding coordinates in the target coordinate system for that occurrence (or vice versa). These coefficients handle variations in translation and angle.

Use the following equations:

$$x_d = A x_s + B y_s + C$$

$$y_d = E x_s + F y_s + D$$

where A, B, C, D, E, and F are the transformation coefficients (forward or reverse); x_s and y_s specify the source coordinates (with respect to the origin of the model coordinate system for a forward transformation or target coordinate system for a reverse transformation); and, x_d and y_d are the destination coordinates (with respect to the origin of the target coordinate system for a forward transformation or model coordinate system for a reverse transformation).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving a transformation coefficient from a find AGM result buffer																
ResultAgmId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultType																
ResultArrayPtr - Possible values returned																
Find AGM result buffer ID for transformation coefficient results	Specifies a find AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_FIND_RESULT, and used to store MagmFind() results.															
M_A_FORWARD +	Retrieves the forward transformation coefficient A for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_A_REVERSE +	Retrieves the reverse transformation coefficient A for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_B_FORWARD +	Retrieves the forward transformation coefficient B for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_B_REVERSE +	Retrieves the reverse transformation coefficient B for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_C_FORWARD +		Retrieves the forward transformation coefficient C for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_C_REVERSE +		Retrieves the reverse transformation coefficient C for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_D_FORWARD +		Retrieves the forward transformation coefficient D for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_D_REVERSE +		Retrieves the reverse transformation coefficient D for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_E_FORWARD +		Retrieves the forward transformation coefficient E for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_E_REVERSE +		Retrieves the reverse transformation coefficient E for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_F_FORWARD +		Retrieves the forward transformation coefficient F for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_F_REVERSE +		Retrieves the reverse transformation coefficient F for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve results for a positive (blue) or negative (red) rectangle from a train AGM result buffer, set ResultType to one of the following values. In this case, set the Index parameter to the index of the positive or negative rectangle.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving rectangle results from a train AGM result buffer																	
ResultAgmId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultType																	
ResultArrayPtr - Possible values returned																	
Train AGM result buffer ID for rectangle-specific results	Specifies a train AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT, and used to store MagmTrain() results.																
M_ANGLE +	Retrieves the angle of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +	Retrieves the X-coordinate of the center of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER_Y +	Retrieves the Y-coordinate of the center of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_TOP_LEFT_X +	Retrieves the X-coordinate of the top-left corner of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_TOP_LEFT_Y +	Retrieves the Y-coordinate of the top-left corner of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_IMAGE_INDEX +	Retrieves the index of the training image that contains the specified rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_HEIGHT +	Retrieves the height of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_WIDTH +	Retrieves the width of the rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve results for a model from a train AGM result buffer, set ResultType to one of the following values. In this case, set the Index parameter to the index of the model.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving model results from a train AGM result buffer																
ResultAgmId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultType																
ResultArrayPtr - Possible values returned																
Train AGM result buffer ID for model-specific results	Specifies a train AGM result buffer, allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT, and used to store MagmTrain() results.															
M_STATUS +	Retrieves the status of the model.	c		e	f	g	a		i		j	k	l	m	n	b

		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_STATUS_TRAIN_AUTO_LABELING_FAILED	Specifies that the automatic labeling of negative model locations was unsuccessful. In this case, the model could not be trained and is unusable. You cannot copy the failed model to a find AGM context using MagmCopyResult().	c		e	f	g	a		i		j	k	l	m	n	b
		M_STATUS_TRAIN_FAILED	Specifies that the model could not be trained and is unusable. You cannot copy the failed model to a find AGM context using MagmCopyResult().	c		e	f	g	a		i		j	k	l	m	n	b
		M_STATUS_TRAIN_OK	Specifies that the model was successfully trained and is usable in a find AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRAIN_DETECTION_ANGLE +		Retrieves whether using rotated labeled image regions (red and blue rectangles) to train the composite-definition model was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
		ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies that the use of rotated labeled image regions was disabled.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies that the use of rotated labeled image regions was enabled.	c		e	f	g	a		i		j	k	l	m	n	b

M_TRAINING_CONFIDENCE +	Retrieves the training confidence of the trained composite-definition model. The training confidence represents an estimation of the difference between the detection scores at the positive positions and the detection scores at the negative positions in the training images. The detection score is the likelihood that there is a true occurrence at the position. Note that a good training confidence is 20.0% or higher. The higher the training confidence, the better the trained composite-definition model will be at discriminating true occurrences from false occurrences. Note that the training confidence should be used to evaluate the trained composite-definition model. If the training confidence is too low, you should adjust your training settings and re-train the composite-definition model. This result type is only available when M_STATUS is M_STATUS_TRAIN_OK and is only meaningful when M_TRAINING_ERROR is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the training confidence.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRAINING_ERROR +	Retrieves the percentage of training image labels that were misclassified by the MagmTrain() operation. For example, assume you train the composite-definition model using images that contain 5 positive (blue) rectangles and 5 negative (red) rectangles. If the detection score at 4/5 of the positive positions is greater than or equal to the trained acceptance level and the detection score at 4/5 of the negative positions is correctly below the trained acceptance level, 2/10 positions are misclassified and the training error will be 20.0%. A training error of 0.0% means that the trained composite-definition model is able to detect all positively labeled positions without incorrectly detecting any negatively labeled positions. This result type is only available when M_STATUS is M_STATUS_TRAIN_OK.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for
Tables:

- For retrieving general results from a find or train AGM result buffer
- For retrieving a result of a single occurrence or all occurrences from a find AGM result buffer
- For retrieving a transformation coefficient from a find AGM result buffer
- For retrieving rectangle results from a train AGM result buffer
- For retrieving model results from a train AGM result buffer

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to determine whether a result is available.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether results are available																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_AVAILABLE	Retrieves whether the requested result type is available for retrieval.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the requested result type is not available.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the requested result type is available.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values of the ResultType parameter listed in For retrieving a result of a single occurrence or all occurrences from a find AGM result buffer, For retrieving a transformation coefficient from a find AGM result buffer.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																	
ResultType combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	
M_NB_ELEMENTS	Retrieves the required array size (number of elements) to store the returned values.																
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For retrieving general results from a find or train AGM result buffer
- For retrieving a result of a single occurrence or all occurrences from a find AGM result buffer
- For retrieving a transformation coefficient from a find AGM result buffer
- For retrieving rectangle results from a train AGM result buffer
- For retrieving model results from a train AGM result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested results to the required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested results to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested results to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested results to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested results to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_DOUBLE*. If the requested information does not

fit into an *AIL_DOUBLE*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmInquire

Synopsis

Inquire information about a specified AGM context, model, or result buffer.

Syntax

```
AIL_INT64 MagmInquire(
    AIL_ID ContextOrResultId, //in
    AIL_INT64 Index,          //in
    AIL_INT64 InquireType,    //in
    void *UserVarPtr          //out
)
```

Description

This function inquires about a setting of an AGM context, a model contained therein, or an AGM result buffer.

Parameters

ContextOrResultId

Specifies the identifier of the find or train AGM context or result buffer about which to inquire information.

See the Parameter associations section for possible values that can be specified.

Index

Specifies that information will be inquired about an AGM context, an individual model, or an AGM result buffer. Set this parameter to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a context, model, or result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default. If an AGM context is specified, same as M_CONTEXT. If an AGM result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_AGM_MODEL_INDEX(AIL_INT ModelIndex)	Inquires information about an individual model.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ModelIndex Set this parameter to the following:															

			0 Specifies the index of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Inquires information about an AGM context.			c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Inquires information about an AGM result buffer.			c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the type of information about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MagmInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ContextOrResultId** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring about a find or train AGM context
- For inquiring about the type of model in a find or train AGM context
- For inquiring about a model in an AGM context
- For inquiring about a find or train AGM result buffer

To inquire information about a find or train AGM context, the InquireType parameter can be set to one of the following values. In this case, you must set the Index parameter to M_CONTEXT or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a find or train AGM context																
ContextOrResultId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

InquireType																
UserVarPtr - Possible values returned																
AGM context ID	Specifies the identifier of a find or train AGM context, allocated using MagmAlloc().															
M_MODIFICATION_COUNT +	Inquires the current value of the modification counter. The modification counter is increased by one each time settings for the context are modified. Although you cannot identify the modification counter's contents, you can compare them throughout your application to know if the context has been altered. If the modification counter has changed you can, for example, prompt the user to save before closing the application.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the current value of the modification counter.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_MODELS +	Inquires the number of models in the AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 1	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_OWNER_SYSTEM +	Inquires the identifier of the system on which the AGM context was allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

	M_PREPROCESSED +	Inquires whether the AGM context is preprocessed. The context must be preprocessed (using MagmPreprocess()) before calling MagmFind() or MagmTrain(). After certain settings of the AGM context are changed with MagmControl(), a model is added or removed with MagmDefine(), or a composite-definition model is copied with MagmCopyResult(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire information about the type of model in a find or train AGM context, the InquireType parameter can be set to the following value. In this case, you must set the Index parameter to the index of a specific model, using M_AGM_MODEL_INDEX().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the type of model in a find or train AGM context																	
ContextOrResultId		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	InquireType																
		UserVarPtr - Possible values returned															
AGM context ID with a model		Specifies a find or train AGM context, allocated using MagmAlloc(). It must contain a model.															
	M_MODEL_TYPE +	Inquires the type of the specified model.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COMPOSITE	Specifies a composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SINGLE	Specifies a single-definition model.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire information about a model in an AGM context, the InquireType parameter can be set to one of the following values. In this case, you must set the Index parameter to the index of a specific model, using M_AGM_MODEL_INDEX().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a model in an AGM context																
ContextOrResultId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																
UserVarPtr - Possible values returned																
Find AGM context ID with a composite-definition model	Specifies a find AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND and used in MagmFind() operations. It must contain a composite-definition model.															
M_ACCEPTANCE_COVERAGE +	Inquires the minimum coverage score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable coverage score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_DETECTION +	Inquires the minimum detection score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable detection score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_DETECTION_FROM_TRAIN +	Inquires the minimum detection score required for an occurrence to be considered a match; this score was automatically determined during the training of the composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum acceptable detection score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_DETECTION_SOURCE +	Inquires whether the trained acceptance level or a custom acceptance level is used for the detection score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_FROM_TRAIN	Specifies to use the detection score's acceptance level established during the training of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the value specified with M_ACCEPTANCE_DETECTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT +		Inquires the minimum fit score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT_OVERALL +		Inquires the minimum overall fit score required for an occurrence to be considered a match. This inquire type is only available when M_MODEL_SOURCE is set to M_USER_IMAGE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable overall fit score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_DELTA_ANGLE_SOURCE +		Inquires whether to use the trained detection angle range or a custom detection angle range to limit the angle of detection. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the detection angle range established during the training of the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the values set with M_DETECTION_DELTA_NEG_ANGLE and M_DETECTION_DELTA_POS_ANGLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DETECTION_DELTA_NEG_ANGLE +	Inquires the lower limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *LowerLimit* = M_DETECTION_REF_ANGLE - M_DETECTION_DELTA_NEG_ANGLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the lower limit of the detection angle range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_DELTA_POS_ANGLE +	Inquires the upper limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *UpperLimit* = M_DETECTION_REF_ANGLE + M_DETECTION_DELTA_POS_ANGLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the upper limit of the detection angle range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_MULTI_ANGLE_MODE +	Inquires the mode with which to detect occurrences when searching at multiple angles. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GLOBAL	Specifies to detect occurrences across all search angles globally.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INDIVIDUAL	Specifies to detect occurrences at each search angle individually.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_REF_ANGLE +	Inquires the nominal search angle; this is the angle at which you expect to find the model's reference axis in the target. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		0.0 <= *Value* < 360.0	Specifies the nominal search angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_STEP_ANGLE +		Inquires the step angle at which to search for occurrences of the model. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		5.0 <= *Value* <= 180.0	Specifies the step angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EMPTY_REGIONS_MODE +		Inquires whether the regions of the trained composite-definition model that do not contain edges are considered when searching for model occurrences. Note that you can set this in the train AGM context using M_EMPTY_REGIONS_MODE before training; once the model is trained and copied to a find AGM context, you cannot change this setting. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically determine the empty regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to consider empty regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_MODE +		Inquires the mode of the fit operation. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to disable the fit operation; the positional results are determined by detection only.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PIXEL_PRECISION	Specifies to calculate the fit with pixel precision.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SUBPIXEL_PRECISION	Specifies to calculate the fit with subpixel precision. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_HAS_USER_MODEL +	Inquires whether the model contains a separate image to use for the fit and coverage calculations when M_MODEL_SOURCE is set to M_USER_IMAGE. For a find AGM context containing a trained composite-definition model, you can specify a model image buffer using MagmControl() with M_MODEL_IMAGE. You can then set M_MODEL_SOURCE to use this model image for the fit and coverage calculations. If a model image buffer was not specified, this inquire type will return M_FALSE, and the trained composite-definition model will be used for these calculations. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the model does not contain a separate image.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the model contains a separate image.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_ACCEPTED_OVERLAP +	Inquires the maximum allowable overlap between found occurrences, as a percentage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the acceptable overlap, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_ASSOCIATION_DISTANCE +	Inquires the maximum distance that a model edgel can be from the nearest occurrence edgel for that edgel to count towards the regular fit operation and increase the coverage score. Note that the maximum association distance has no effect on the overall fit score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
1.0 <= *Value* <= 4.0	Specifies the maximum association distance, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_PRECISION +	Inquires the percentage of the trained composite-definition model's edgels to use when searching for occurrences. Note that you can set this in the train AGM context using M_MODEL_PRECISION before training; once the model is trained and copied to a find AGM context, you cannot change this setting. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the amount of model edgels, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_SOURCE +	Inquires whether the trained composite-definition model or a prototypical image of the model is used for the fit and coverage calculations. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FROM_TRAIN	Specifies to use the trained composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_IMAGE	Specifies to use the image set with M_MODEL_IMAGE.	c		e	f	g	a		i		j	k	l	m	n	b
M_PERSEVERANCE_DETECTION +	Inquires the algorithm's perseverance when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the perseverance, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_X +	Inquires the X-coordinate of the origin of the model's reference axis, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_REFERENCE_Y +	Inquires the Y-coordinate of the origin of the model's reference axis, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Inquires the width of the model. The width of the trained composite-definition model is the same width as the training images' rectangles.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 6	Specifies the width of the model, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Inquires the height of the model. The height of the trained composite-definition model is the same height as the training images' rectangles.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 6	Specifies the height of the model, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_SMOOTHNESS +	Inquires the degree of smoothness (noise reduction) applied to the model image and target image(s) during the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_SMOOTHNESS_FROM_TRAIN +	Inquires the degree of smoothness (noise reduction) applied to the training images during the edge extraction. Note that you can set this in the train AGM context using M_SMOOTHNESS before training; once the model is trained and copied to a find AGM context, you cannot change this setting. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS_SOURCE +	Inquires whether the smoothness value set during training or a custom smoothness value is used for the degree of noise reduction applied to the model image and target image(s) during the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the smoothness value set during the training of the model. This value was set with M_SMOOTHNESS in a train AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the smoothness value set with M_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_CANDIDATES_SCORE +	Inquires the type of score used to sort candidates. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_COVERAGE	Specifies to use the coverage score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_DETECTION	Specifies to use the detection score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_FIT	Specifies to use the fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b

		M_SCORE_FIT_OVERALL	Specifies to use the overall fit score to sort candidates. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STABLE_EDGELS_MODE +		Inquires whether only stable edges are used when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to use only stable edgels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SMOOTHNESS_STABLE	Specifies to use only stable edgels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_X +		Inquires the X step size of explored positions to search for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1	Specifies the X step size.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_Y +		Inquires the Y step size of explored positions to search for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1	Specifies the Y step size.	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE +		Inquires the threshold mode of the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies no threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies a high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_VERY_HIGH	Specifies a very high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE_FROM_TRAIN +		Inquires the threshold mode of the edge extraction used during training. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies no threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies a high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE_SOURCE +		Inquires whether the threshold mode set during training or a custom threshold mode is used for the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FROM_TRAIN	Specifies to use the threshold mode set during the training of the model. This value was set with M_THRESHOLD_MODE in a train AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the threshold mode set with M_THRESHOLD_MODE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRAIN_DETECTION_ANGLE +		Inquires whether using rotated labeled image regions (red and blue rectangles) to train the composite-definition model was enabled. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_DISABLE	Specifies that the use of rotated labeled image regions was disabled.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies that the use of rotated labeled image regions was enabled.	c		e	f	g	a		i		j	k	l	m	n	b
Find AGM context ID with a single-definition model			Specifies a find AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND and used in MagmFind() operations. It must contain a single-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_COVERAGE +		Inquires the minimum coverage score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable coverage score, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_DETECTION +		Inquires the minimum detection score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable detection score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT +		Inquires the minimum fit score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable fit score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCEPTANCE_FIT_OVERALL +		Inquires the minimum overall fit score required for an occurrence to be considered a match. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		0.0 <= *Value* <= 100.0	Specifies the minimum acceptable overall fit score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_MARGIN_BOTTOM +		Inquires the margin at the bottom of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_MARGIN_LEFT +		Inquires the margin at the left of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_MARGIN_RIGHT +		Inquires the margin at the right of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

	M_BOX_MARGIN_TOP +	Inquires the margin at the top of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 163840000.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_OFFSET_X +	Inquires the X-offset of the top-left corner of the model box from the model origin, if the model was defined from a CAD DXF file. For M_DXF_FILE models, the origin is the same as the one defined in the CAD DXF file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... M_BOX_OFFSET_X is updated when the margins that define the model box are changed. This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-offset, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_OFFSET_Y +	Inquires the Y-offset of the top-left corner of the model box from the model origin, if the model was defined from a CAD DXF file. For M_DXF_FILE models, the origin is the same as the one defined in the CAD DXF file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... M_BOX_OFFSET_Y is updated when the margins that define the model box are changed. This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-offset, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_SIZE_X +	Inquires the size along the X-axis of the model box, if the model was defined from a CAD DXF file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_SIZE_Y +	Inquires the size along the Y-axis of the model box, if the model was defined from a CAD DXF file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		Value	Specifies the Y-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CAD_Y_AXIS +		Inquires the direction of the Y-axis for a model of type M_DXF_FILE. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_FLIP	Specifies to flip the Y-axis for the model so that the Y-axis is positive going down.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NO_FLIP	Specifies not to flip the Y-axis for the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_DELTA_NEG_ANGLE +		Inquires the lower limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *LowerLimit* = M_DETECTION_REF_ANGLE - M_DETECTION_DELTA_NEG_ANGLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the lower limit of the detection angle range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_DELTA_POS_ANGLE +		Inquires the upper limit of the detection angle range, relative to the nominal search angle (M_DETECTION_REF_ANGLE). That is, *UpperLimit* = M_DETECTION_REF_ANGLE + M_DETECTION_DELTA_POS_ANGLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 360.0	Specifies the upper limit of the detection angle range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_MULTI_ANGLE_MODE +		Inquires the mode with which to detect occurrences when searching at multiple angles. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GLOBAL	Specifies to detect occurrences across all search angles globally.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INDIVIDUAL	Specifies to detect occurrences at each search angle individually.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_REF_ANGLE +		Inquires the nominal search angle; this is the angle at which you expect to find the model's reference axis in the target. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* < 360.0	Specifies the nominal search angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DETECTION_STEP_ANGLE +		Inquires the step angle at which to search for occurrences of the model. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		5.0 <= *Value* <= 180.0	Specifies the step angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGEL_ANGLE_MODE +		Inquires whether edgel orientation, defined by a gradient angle, is considered when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to consider edgel orientation.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ORIENTATION	Specifies to consider edgel orientation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGES_BOX_SIZE_X +		Inquires the size along the X-axis of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGES_BOX_SIZE_Y +		Inquires the size along the Y-axis of the bounding box of the model's active edges, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EMPTY_REGIONS_MODE +		Inquires whether the regions of the model that do not contain edges are considered when searching for model occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically determine the empty regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to consider empty regions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIRST_LEVEL +		Inquires the resolution level for the initial stage of the search when M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <= 7	Specifies the resolution level. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIRST_LEVEL_USED +	Inquires the resolution level used for the initial stage of the search. When M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED, this value is equal to M_FIRST_LEVEL. When M_PYRAMID_LEVELS_MODE is set to M_AUTO_CONTENT_BASED, this value is the first level that was automatically determined during preprocessing. Note, this value is only available after preprocessing.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <= 7	Specifies the resolution level used.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIT_MODE +	Inquires the mode of the fit operation. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to disable the fit operation; the positional results are determined by detection only.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PIXEL_PRECISION	Specifies to calculate the fit with pixel precision.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SUBPIXEL_PRECISION	Specifies to calculate the fit with subpixel precision. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LAST_LEVEL +	Inquires the resolution level for the final stage of the search when M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		0 <= *Value* <= 7	Specifies the resolution level. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_LAST_LEVEL_USED +	Inquires the resolution level used for the final stage of the search. When M_PYRAMID_LEVELS_MODE is set to M_USER_DEFINED, this value is equal to M_LAST_LEVEL. When M_PYRAMID_LEVELS_MODE is set to M_AUTO_CONTENT_BASED, this value is the last level that was automatically determined during preprocessing. Note, this value is only available after preprocessing.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 7	Specifies the resolution level used.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_ACCEPTED_OVERLAP +	Inquires the maximum allowable overlap between found occurrences, as a percentage. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the acceptable overlap, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_ASSOCIATION_DISTANCE +	Inquires the maximum distance that a model edgel can be from the nearest occurrence edgel for that edgel to count towards the regular fit operation and increase the coverage score. Note that the maximum association distance has no effect on the overall fit score. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
1.0 <= *Value* <= 4.0	Specifies the maximum association distance, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_PRECISION +	Inquires the percentage of model edgels to use when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the amount of model edgels, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_PERSEVERANCE_DETECTION +	Inquires the algorithm's perseverance when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the perseverance, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL_SCALE +	Inquires the pixel scale of the model, if the model was defined from a CAD DXF file. This inquire type is only available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 10000.0	Specifies the pixel scale. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_POLARITY +	Inquires the expected polarity of occurrences, compared to that of the model. This inquire type is not available for models of type M_DXF_FILE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the occurrences can be any polarity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the polarity of occurrences is the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the polarity of occurrences is the same as that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PYRAMID_LEVELS_MODE +	Inquires how to establish the resolution levels for the search. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_AUTO_CONTENT_BASED	Specifies to automatically determine the levels according to the model content.	c		e	f	g	a		i		j	k	l	m	n	b
		M_USER_DEFINED	Specifies to use the values set with M_FIRST_LEVEL and M_LAST_LEVEL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_X +		Inquires the X-coordinate of the origin of the model's reference axis, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-coordinate, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_REFERENCE_Y +		Inquires the Y-coordinate of the origin of the model's reference axis, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-coordinate, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SMOOTHNESS +		Inquires the degree of smoothness (noise reduction) applied to the model image and target image(s) during the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SORT_CANDIDATES_SCORE +		Inquires the type of score used to sort candidates. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_COVERAGE	Specifies to use the coverage score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b

		M_SCORE_DETECTION	Specifies to use the detection score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_FIT	Specifies to use the fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SCORE_FIT_OVERALL	Specifies to use the overall fit score to sort candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STABLE_EDGELS_MODE +		Inquires whether only stable edges are used when searching for occurrences. This inquire type is only available for models of type M_IMAGE and M_IMAGE_REGION. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to use only stable edgels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SMOOTHNESS_STABLE	Specifies to use only stable edgels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_X +		Inquires the X step size of explored positions to search for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1	Specifies the X step size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_Y +		Inquires the Y step size of explored positions to search for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value >= 1	Specifies the Y step size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THRESHOLD_MODE +		Inquires the threshold mode of the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_DISABLE	Specifies no threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_HIGH	Specifies a high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LOW	Specifies a low threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MEDIUM	Specifies a medium threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_VERY_HIGH	Specifies a very high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_USE_MAGNITUDE_TARGET +		Inquires whether to use the gradient magnitude or to use extracted edges when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to extract and use edges.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use the magnitude of the gradient.	c		e	f	g	a		i		j	k	l	m	n	b
Train AGM context ID with a composite-definition model			Specifies a train AGM context, allocated using MagmAlloc() with M_GLOBAL_EDGE_BASED_TRAIN and used in MagmTrain() operations. It must contain a composite-definition model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EMPTY_REGIONS_MODE +		Inquires whether the regions of the model appearances that do not contain edges are considered when training the composite-definition model. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_AUTO	Specifies to automatically determine the empty regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to consider empty regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_PRECISION +	Inquires the percentage of the trained composite-definition model's edgels to use when searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the amount of model edgels, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEGATIVE_LABELS_MAX_OVERLAP +	Inquires the maximum allowable overlap between two automatically generated negative (red) rectangles. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 90.0	Specifies the maximum allowable overlap, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEGATIVE_LABELS_MODE +	Inquires whether to automatically label negative model locations in the training images or to use manually defined labels. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCORDING_TO_LABELS	Specifies to use manually defined negative (red) rectangles if there is at least one red rectangle in the set of training images; otherwise, specifies to automatically label the negative model locations.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically label the negative model locations. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use manually defined negative (red) rectangles. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEGATIVE_LABELS_POSITIVE_MAX_OVERLAP +	Inquires the maximum allowable overlap between an automatically generated negative (red) rectangle and a positive (blue) rectangle. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* < 80.0	Specifies the maximum allowable overlap, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b

M_REALIGN_STRENGTH +	Inquires the strength with which to displace the blue rectangles, which define the positive model locations in the training images, to better align them. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the strength of the realignment.	c		e	f	g	a		i		j	k	l	m	n	b
M_SMOOTHNESS +	Inquires the degree of smoothness (noise reduction) applied to the training images during the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_MODE +	Inquires the threshold mode of the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies no threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_HIGH	Specifies a high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LOW	Specifies a low threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MEDIUM	Specifies a medium threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_VERY_HIGH	Specifies a very high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TRAIN_DETECTION_ANGLE +	Inquires whether to train the composite-definition model using rotated labeled image regions (red and blue rectangles). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_ACCORDING_TO_LABELS	Specifies to use rotated labeled image regions if there is at least one rotated labeled image region in the set of training images; otherwise, specifies not to use rotated labeled image regions.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to use rotated labeled image regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use rotated labeled image regions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRAIN_LABELS_NOISE_MODE +		Inquires the mode with which to handle substandard labeled image regions when training the composite-definition model. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to handle substandard labeled image regions differently. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LABELS_CONFIDENCE	Specifies to be robust to substandard labels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_USE_MAGNITUDE_TARGET +		Inquires whether to use the gradient magnitude or to use extracted edges when training the composite-definition model and searching for occurrences. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies to extract and use edges.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to use the magnitude of the gradient.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire information about a find or train AGM result buffer, the InquireType parameter can be set to one of the following values. In this case, you must set the Index parameter to M_GENERAL or M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a find or train AGM result buffer																		
ContextOrResultId		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	InquireType																	

		UserVarPtr - Possible values returned															
AGM result buffer ID			Specifies the identifier of a find or train AGM result buffer, allocated using MagmAllocResult().														
	M_MODIFICATION_COUNT +		Inquires the current value of the modification counter. The modification counter is increased by one each time that you call MagmFind() or MagmTrain(). Although you cannot identify the modification counter's contents, you can compare them throughout your application to know if the result buffer has been altered.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the current value of the modification counter.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OWNER_SYSTEM +		Inquires the identifier of the system on which the AGM result buffer was allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
		System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a find or train AGM context
- For inquiring about the type of model in a find or train AGM context
- For inquiring about a model in an AGM context
- For inquiring about a find or train AGM result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a find or train AGM context
- For inquiring about the type of model in a find or train AGM context
- For inquiring about a model in an AGM context
- For inquiring about a find or train AGM result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to determine whether an inquire type is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To inquire whether an inquire type is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b

M_SUPPORTED	Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a find or train AGM context
- For inquiring about the type of model in a find or train AGM context
- For inquiring about a model in an AGM context
- For inquiring about a find or train AGM result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To specify the data type																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*. Note that M_TYPE_AIL_ID should only be used with M_OWNER_SYSTEM.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b

M_TYPE_AIL_INT64		Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmPreprocess

Synopsis

Preprocess an AGM context.

Syntax

```
void MagmPreprocess(
    AIL_ID ContextAgmId,  //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function preprocesses the specified AGM context. It sets internal settings so that future search or train operations will be optimized for speed and robustness. The procedure is potentially quite lengthy (up to a few seconds) and increases the size of the AGM context significantly.

This function must be called before performing an MagmFind() or MagmTrain() operation. Call this function after all the required settings have been set; if some of the model's settings are changed after a call to **MagmPreprocess**(), the model must be preprocessed again. Note that you must preprocess the context after copying a composite-definition model to a find AGM context, even if no settings are changed. When you save the AGM context, the model's preprocessing changes are not stored with the model. Upon restoration, the model must be preprocessed again.

Parameters

ContextAgmId

Specifies the AGM context to preprocess. The AGM context must have been previously allocated on the system using MagmAlloc() with M_GLOBAL_EDGE_BASED_FIND or M_GLOBAL_EDGE_BASED_TRAIN.

ControlFlag

Specifies whether to preprocess the AGM context or remove preprocessing information from the AGM context. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to perform preprocessing																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Performs the preprocessing of the AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESET	Removes preprocessing information from the AGM context. Removing this information can be useful if you want to conserve computer memory within an application and preserve AGM context settings.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.

Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmRestore

Synopsis

Restore an AGM context from disk.

Syntax

```
AIL_ID MagmRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *ContextAgmPtr        //out
)
```

Description

This function restores an AGM context that was previously saved to a file, using MagmSave() or MagmStream(). This function restores all of the AGM context's settings that were in effect when the AGM context was saved. A restored AGM context is not preprocessed, therefore you must call MagmPreprocess() before calling MagmFind() or MagmTrain().

When the restored AGM context is no longer required, release it using MagmFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the restored AGM context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the AGM context. The function handles (internally) the opening and closing of the file.

This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, AGM context files have an MAGM extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the AGM context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

ContextAgmPtr

Specifies the address of the variable in which to write the AGM context identifier or specifies the data type that the function should use to return the AGM context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated AGM context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_AGM_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the AGM context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find AGM context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the restored find AGM context. If restoration fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the train AGM context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the restored train AGM context. If restoration fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the AGM context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_AGM_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.

DLL	Requires ail11.dll; ailagm11.dll.

MagmSave

Synopsis

Save an AGM context to a file.

Syntax

```
void MagmSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID ContextId,            //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about the previously allocated AGM context to disk, including the model's current settings. This information can be reloaded, using MagmRestore() or MagmStream(). However, preprocessing changes are not saved.

Parameters

FileName

Specifies the name and path of the file in which to save the AGM context; it is recommended that you use the MAGM file extension for easier use with other Aurora Imaging software products. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, AGM context files have an MAGM extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

ContextId

Specifies the identifier of the AGM context to save.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmStream

Synopsis

Load, restore, or save an AGM context from/to a file or a memory stream.

Syntax

```
void MagmStream(
    AIL_TEXT_PTR MemPtrOrFileName, //in-out
    AIL_ID SysId,                  //in
    AIL_INT64 Operation,           //in
    AIL_INT64 StreamType,          //in
    AIL_DOUBLE Version,            //in
    AIL_INT64 ControlFlag,         //in
    AIL_ID *ContextAgmIdPtr,       //in-out
    AIL_INT *SizeByteVarPtr        //out
)
```

Description

This function can load, restore, or save an AGM context from/to a file or memory stream.

To inquire the number of bytes necessary to save an AGM context to memory stream, you should first call this function (**MagmStream**()) with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. This function is equivalent to a file saved using MagmSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file. All information about the previously allocated AGM context is saved, including the individual models current settings. However, preprocessing changes are not saved.

All of the AGM context settings that were in effect when the AGM context was saved will be restored. A loaded or restored AGM context is not preprocessed; therefore, you must call MagmPreprocess() before calling MagmFind() or MagmTrain().

Using **MagmStream**(), you can choose to save a backwards-compatible version of the AGM context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore AGM contexts saved using any Aurora Imaging Library version with which this function was available. Settings that do not exist in the lower version will be filled with default values when the AGM context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the AGM context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform on the AGM context.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the AGM context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of stream																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the AGM context.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

ContextAgmIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the AGM context.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the AGM context, in bytes.

If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of an AGM context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, and **ContextAgmIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.																	
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
MemPtrOrFileName - Possible values passed or returned																
SysId																
Version																
ContextAgmIdPtr - Possible values passed or returned																
M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save an AGM context to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.															
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the AGM context must be compatible when saved. The size of an AGM context might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
ContextAgmIdPtr	Specifies the address of the variable containing the identifier of the AGM context.	c		e	f	g	a		i		j	k	l	m	n	b

M_LOAD	Loads the content of a specified file or memory stream into a previously allocated AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to load the context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. AGM contexts typically have an MAGM file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ContextAgmIdPtr	Specifies the address of the variable containing the identifier of the AGM context. All previous settings are overwritten with those of the loaded context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b

M_RESTORE	Restores an AGM context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to restore the context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. AGM contexts typically have an MAGM file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	Specifies the system on which to allocate the AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b

ContextAgmIdPtr	Specifies the address of the variable in which to write the identifier of the AGM context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE	Saves an AGM context to a specified file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging software products, when saving an AGM context to a file, use the MAGM file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire object. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the AGM context must be compatible when saved.	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
	ContextAgmIdPtr		Specifies the address of the variable containing the identifier of the AGM context to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.

MagmTrain

Synopsis

Train a composite-definition model that is in a train AGM context.

Syntax

```
void MagmTrain(
    AIL_ID TrainContextId,       //in
    const AIL_ID *ImageArrayPtr, //in
    AIL_INT NumImages,           //in
    AIL_ID TrainResultId,        //out
    AIL_INT64 ControlFlag        //in
)
```

Description

This function trains a composite-definition model that is in a train AGM context and stores the result in a train AGM result buffer. The trained composite-definition model is constructed from training images with regions labeled as positive or negative. During training, the model learns what is a favorable feature of an occurrence and what is not. Retrieve results about the training operation using MagmGetResult(). Note, the train AGM context must be preprocessed (using MagmPreprocess()) before calling this function. Once trained, you copy the resulting trained composite-definition model to a find AGM context using MagmCopyResult().

The training process involves using model appearances and background samples to construct a model that is good at finding your model, especially in cluttered target images. These areas must be labeled in the training images using regions of interest (ROIs) containing blue/red rectangles. Blue rectangles define positive model locations, while red rectangles define negative model locations. Negative model locations include clutter in the background and areas in an image that resemble your model but aren't.

You should use blue rectangles to specify appearances of the model that you want to find. **MagmTrain**() can only train one model; therefore, each blue rectangle must identify an occurrence of the same model. Note that model appearances don't need to be in the same location in the MagmFind() target images.

If you enabled the automatic labeling of negative model locations (MagmControl() with M_NEGATIVE_LABELS_MODE set to M_AUTO), you must only label the positive model appearances; **MagmTrain**() will automatically identify and label the background areas of your training images. In this case, a minimum of one training image must have a blue ROI associated with it. The ROI can be composed of several non-contiguous and/or overlapping areas, but it must contain at least one blue rectangle and cannot contain any red rectangles. Note that there can be training images with no ROI (no rectangles).

If you did not enable the automatic labeling of negative model locations, you must manually label both the model appearances and background samples before calling **MagmTrain**(). In this case, the set of training images must contain red and blue rectangles, associated to the images using ROIs. You should use red rectangles to specify clutter in the background. If there is an area in an image that resembles your model but isn't, you can use red rectangles to prevent MagmFind() from returning that area as a possible occurrence. A minimum of one training image must have an ROI associated with it. The ROI can be composed of several non-contiguous and/or overlapping areas, but it must contain at least one red rectangle and/or one blue rectangle. Considering all the rectangles across all training images, there must be a minimum of one red rectangle and one blue rectangle. Note that there can be training images with only red rectangles, only blue rectangles, or no rectangles. The training images with no rectangles are not used during training.

Note that all rectangles must be the same size.

You can use the *AgmLabelingTool* example, accessible from the Example Launcher, to define images with appropriate regions of interest.

Use MagmControl() to configure individual model settings before calling this function. For example, you can set M_EMPTY_REGIONS_MODE to M_AUTO to take into consideration, when training the model, the regions of the model appearances that do not contain edges.

You must call this function before calling MagmCopyResult(). You cannot copy the model to a find AGM context unless it is successfully trained. To check the training status of the model, use MagmGetResult() with M_STATUS.

Parameters

TrainContextId

Specifies the identifier of the train AGM context with which to train the composite-definition model. The train AGM context must have been previously allocated on the required system using MagmAlloc() with M_GLOBAL_EDGE_BASED_TRAIN. It must also contain a composite-definition model, added using MagmDefine() with M_COMPOSITE.

ImageArrayPtr
Accepts the address of one of the following:

- array of type AIL_ID [optionally, in C++: a reference to a constant std::vector<AIL_ID>] (and a maximum array rank for C# of 1)
  Required array size:(NumImages)
- AIL_ID

Specifies either the address of the array containing the buffer identifiers of the training images or specifies the address of the variable containing the identifier of the image buffer container. The image buffers must be 1-band, 8-bit unsigned buffers. The minimum and maximum training image sizes are 6x6 and 32768x32768 pixels, respectively.

A minimum of one image buffer must have a region of interest (ROI) with at least one blue rectangle, defined using MbufSetRegion(); blue rectangles should indicate positive model locations. In addition, if M_NEGATIVE_LABELS_MODE is set to M_USER_DEFINED, at least one image buffer must have an ROI containing a red rectangle; red rectangles should indicate negative model locations. The minimum and maximum rectangle sizes are 6x6 and 32768x32768 pixels, respectively. Note that all rectangles must be the same size and the center of each rectangle must be inside the training image. If M_TRAIN_DETECTION_ANGLE is set to M_DISABLE, all rectangles must have an angle of 0.0°. The region must be defined in vector format from a 2D graphics list (M_VECTOR or M_VECTOR_AND_RASTER). An error will be generated if the ROI is only in raster format (M_RASTER).

NumImages

Specifies the size of the specified image buffer array. When an image buffer container is specified, you must set NumImages to 1.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the ImageArrayPtr parameter.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the number of training images

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 1	Specifies the size of the array passed to ImageArrayPtr.	c		e	f	g	a		i		j	k	l	m	n	b

TrainResultId

Specifies the train AGM result buffer identifier. The result buffer must have been allocated using MagmAllocResult() with M_GLOBAL_EDGE_BASED_TRAIN_RESULT.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailagm11.lib.
DLL	Requires ail11.dll; ailagm11.dll.